

Structure transfer and consolidation in visual implicit learning

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This study investigates the conditions under which abstract knowledge transfers to new learning. It presents **solid** evidence across a number of behavioral experiments that when explicit awareness of learned statistical structure is present, knowledge can transfer immediately, but that otherwise similar transfer requires sleep-dependent consolidation. The **valuable** results provide new constraints on theories of transfer learning and consolidation.

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Abstract

Transfer learning, the re-application of previously learned higher-level regularities to novel input, is a key challenge in cognition. While previous empirical studies investigated human transfer learning in supervised or reinforcement learning for explicit knowledge, it is unknown whether such transfer occurs during naturally more common implicit and unsupervised learning and, if so, how it is related to memory consolidation. We compared the transfer of newly acquired explicit and implicit abstract knowledge during unsupervised learning by extending a visual statistical learning paradigm to a transfer learning context. We found transfer during unsupervised learning but with important differences depending on the explicitness/implicitness of the acquired knowledge. Observers acquiring explicit knowledge during initial learning could transfer the learned structures immediately. In contrast, observers with the same amount but implicit knowledge showed the opposite effect, a structural interference during transfer. However, with sleep between the learning phases, implicit observers, while still remaining implicit, switched their behaviour and showed the same pattern of transfer as explicit observers did. This effect was specific to sleep and not found after non-sleep consolidation. Our results highlight similarities and differences between explicit and implicit learning while acquiring generalizable higher-level knowledge and relying on consolidation for restructuring internal representations.

Introduction

Recent years have seen a rise in interest in *transfer learning*, the application of previously learned regularities to novel input, as the effectiveness of this process is a key feature for biological and artificial intelligence alike (1–3). An important factor for successful transfer is the formation of abstract, generalizable knowledge transcending simple representation of the specific surface properties of the input (4–7). A challenging aspect of this memory formation is balancing the search for new commonalities between different tasks or contexts and minimizing the interference between new experiences and old memories (8–10). Previous empirical studies on transfer learning focused on the domains of supervised or reinforcement learning and mainly dealt with explicit knowledge. Although this allows for straightforward designs and ensures continuity with previous research, it ignores the fact that a large part of ecologically relevant learning happens implicitly (11) and in the absence of supervision or reinforcement (12, 13). It is, therefore, an open question whether and how humans' transfer learning operates during unsupervised learning and in the absence of explicit knowledge.

In the current study, we contrast implicit and explicit transfer learning in an unsupervised learning paradigm. Directly contrasting implicit and explicit learning is important, as previous studies demonstrated that explicit and implicit forms of learning can show different outcomes and learning trajectories (14–19). In addition, we investigated the effect of consolidation on implicit transfer learning at various timescales since the role of consolidation in structural abstraction and transfer has been demonstrated previously for explicit learning with supervision or feedback (20–25) and since there exist interactions between the level of explicitness and consolidation (26–30). Our paradigm builds on the classical spatial visual statistical learning (SVSL) design (31) and extends it to a transfer learning setup. In this new setup, participants proceed from learning reappearing patterns embedded in unsegmented input to abstracting their shared structure and finally applying this newly abstracted structure to novel unsegmented input. This paradigm, therefore, combines the challenge of segmentation based on co-occurrence statistics (statistical learning) with the abstraction and reapplication of shared structures (transfer learning), setting our study apart from previous studies on implicit abstraction or generalization (32–34).

Our results show that during unsupervised learning, the transfer of learned structures is immediately possible when explicit knowledge is acquired. In contrast, when only implicit knowledge is obtained, immediate transfer leads to an opposite effect, an interference between old knowledge and new learning. However, after 12-hour sleep consolidation following the learning of the first structure, while the knowledge still remains implicit, the same successful transfer of learned structures (i.e., generalization) occurs as in the explicit condition. Hence, our results extend the empirical findings on transfer learning to the domain of unsupervised implicit learning by showing that transfer learning is possible based on both explicit and implicit knowledge, albeit with different trajectories, and highlight two important causes for these different trajectories: explicitness and consolidation of the knowledge to be transferred.

Results

Visual Statistical Learning as a Testbed for Unsupervised Hierarchical Structure Learning

Our experimental setup builds on the standard spatial visual statistical learning framework (SVSL) (31) and extends it to a transfer learning paradigm (Figure 1). In the traditional SVSL, participants passively watch a stream of scenes, each consisting of multiple shapes on a grid layout, which, unbeknown to the participant, are composed of a set of fixed shape pairs. The pairs are “fixed” since the two constituent shapes of a pair always appear next to each other in the scenes in a fixed spatial relationship. Importantly, each pair abuts at least one other pair in the scenes, so the pairs are not separated visually from each other. Thus, the identities of the shape pairs need to be extracted by learning across multiple scenes to achieve successful segmentation. Adult and infant humans, as well as various other species, have been shown to automatically and implicitly extract the underlying pair structure of such scenes (35–37).

This paradigm is an ideal candidate for investigating unsupervised learning of higher-order structures as it not only allows learning the fixed pairs of shapes (chunks), but through these chunks, more abstract underlying structures can also be defined. One example of such a more abstract feature is the mean orientation of the learned chunks, where the orientation of a chunk is defined by whether the arrangement of the shapes within the pair is horizontal or vertical. Critically, these more abstract features are not observable properties of the scenes *per se*, as there are no segmentation cues revealing the chunks instantaneously. Thus, only when the chunks are learned can their orientation emerge as a feature. Hence, the latent property of orientation can be extracted only in conjunction with learning the observable pair associations. We used this feature in our transfer learning version of the SVSL paradigm, but we went beyond simply assessing whether a previously presented latent structure, such as the mean orientation of the learned chunks, became accessible to the observers. Instead, we measured how exposure to one type of abstract structure differentially influenced the acquisition of multiple types of structures later on.

Specifically, in the first training phase (Phase 1) in all of our experiments conducted online (Methods), participants saw scenes composed from only horizontal pairs or only vertical pairs, depending on the assigned condition counterbalanced across participants. In a second training phase (Phase 2), following some delay, they saw scenes composed from both horizontal and vertical pairs and made from a novel set of shapes not used in Phase 1. In the test phase after Phase 2, participants completed a 2-alternative forced-choice (2AFC) familiarity test indicating in each trial, whether they found more familiar a presented real pair (used in the training phases) or a foil pair (generated by randomly combining two shapes used in the training phase) (Figure 1). Due to a consistency effect detected in the responses for Phase 2 pairs, we used only half of the presented test trials of that Phase for our analysis (see details in the Method section). Our main question was how unsupervised learning of instances of a latent structure (e.g., horizontality by seeing only horizontal chunks) in Phase 1 influenced the unsupervised learning of instances of the same (horizontal) and a novel (vertical) structure in Phase 2. Our main measure of interest was the difference in learning the same vs. different types of pairs during Phase 2.

The explicitness of the acquired knowledge was assessed by using an exit survey at the very end of the experiment (see Supplementary Materials). Participants were labeled as “explicit” if they gave any indication of being aware of reappearing fixed patterns/pairs in the scenes. It is worth noting that we excluded from our analysis the small number of participants who clearly reported awareness of the underlying structure (horizontality/verticality). Therefore, the participants we labeled “explicit” in this study also had only far-from-perfect, partial knowledge about the structure of the scenes. In agreement with earlier visual statistical learning reports, the proportion of these excluded participants was 1–1.5% across experiments, and their exclusion did not change any of our results.

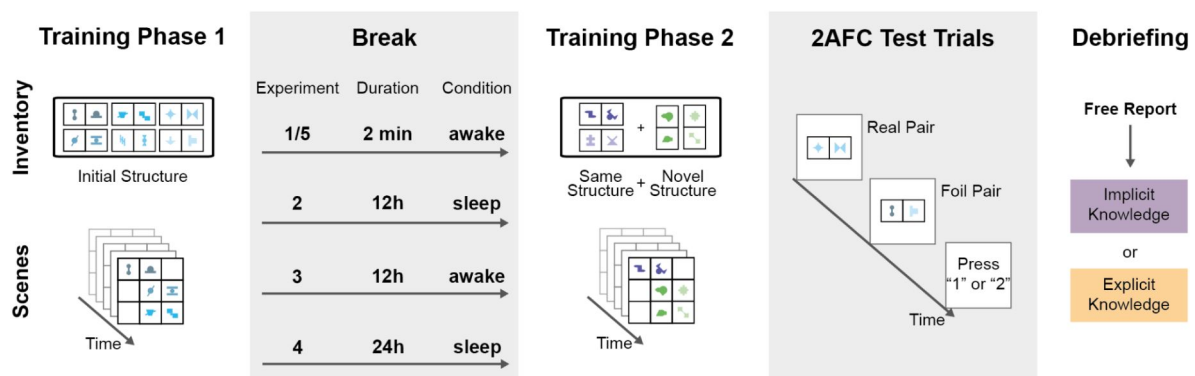


Figure 1.

Overview of the experimental setup. **Training Phase 1:** Participants passively observed a stream of scenes made up of abstract shapes (*lower panel*). Unbeknown to the participants, shapes in the scene appeared only in pairs of fixed spatial configurations defined by the Inventory (*upper panel*). All pairs in Phase 1 had the same underlying structure of either horizontal or vertical orientation. The colors in the figure are only for illustration purposes; for the participants, all shapes were black. **Break:** After Phase 1, there was a break that varied across the five experiments between two minutes and 24 hours. Participants spent the break either in asleep or awake condition. **Training Phase 2:** After the break, the participants were exposed to visual scenes made of a different set of abstract shapes. Half of the created pairs of the new inventory had horizontal, while the other half had vertical underlying structures. **2AFC Test Trials:** After Phase 2, participants completed a series of 2AFC Test Trials, in which they had to decide if a real pair from the training phases or a foil pair, created by random combination of the shapes, was more familiar. **Debriefing:** Finally, participants answered open-ended questions about the experiment, which were used to assess whether they gained explicit knowledge about the presence of shape pairs.

Explicit Learners Generalize, while Implicit Learners Show a Structural Novelty Effect

Experiment 1 ($n=226$ after exclusions; see Methods for details) investigated the immediate transfer of knowledge about the higher level structure between the two learning contexts by implementing a short 2-minute break between Phase 1 and 2 and, thereby, providing a baseline for the subsequent experiments. Both explicit and implicit learners in Experiment 1 performed above chance for pairs of the first training phase (Expl: $M=67.9$, $SE=4.6$, $d=0.67$, $t(33)=3.93$ $p=0.002$, $BF=70.8$; Impl: $M=55.0$, $SE=1.1$, $d=0.32$, $t(191)=4.46$ $p<0.001$, $BF=919$). In addition, the explicit participants ($n=34$) performed significantly better than the implicit ones ($n=192$) ($d=0.73$, $t(224)=2.75$ $p=0.009$) (Figure 2). For Phase 2 pairs, explicit participants performed above chance for learning pairs that shared their higher level orientation structure with that of pairs in Phase 1 ($M=67.6$, $SE=6.3$, $d=0.48$, $t(33)=2.81$ $p=0.033$, $BF=5$). The same participants showed some moderate learning of the new pairs with the novel (non-matching) structure as well, but this learning failed to reach significance ($M=58.1$, $SE=5.6$, $d=0.25$, $t(33)=1.46$ $p=0.465$, $BF=0.48$). In contrast to the explicit participants, implicit participants showed the opposite pattern, performing significantly above chance for pairs of the novel structure ($M=57.8$, $SE=1.9$, $d=0.30$, $t(191)=4.16$ $p<0.001$, $BF=281$) but demonstrated strong evidence that they did not learn pairs sharing the higher level orientation structure with pairs in Phase 1 ($M=49.1$, $SE=2.1$, $d=0.03$, $t(191)=-0.44$ $p=0.659$, $BF=0.09$). The difference between the implicit participants' performances with same vs. different higher-order structures was statistically significant ($M_{diff}=8.72$, $d=0.20$, $t(191)=2.78$ $p=0.030$, $BF=3.38$). This qualitative pattern of the implicit participants' markedly better performance with novel structures was distinctively different from the pattern shown by the explicit participants, as indicated by the significant interaction between the two factors (participant type and structure type) of the two-factor mixed ANOVA results ($F(1,224)=4.89$, $p=0.028$, $BF=31.6$, $\eta_p^2=0.02$). These results imply that participants with more explicit acquired knowledge in Phase 1 effectively generalized the higher-level structure of this learned knowledge to novel situations and contexts, showing a "structure transfer" effect. Meanwhile, implicit participants showed a "structural novelty" effect that might be explained by a structure-level interference due to a larger representational overlap between previously learned and presently seen horizontal pairs than between previous horizontal and present vertical pairs.

To strengthen the reliability of our results, we conducted a conceptual replication of Experiment 1 using two types of diagonal pairs, which were again orthogonal to each other. The results again showed a structural novelty effect for implicit participants (see Supplementary Experiment 1).

Consolidation Enables Implicit Learners to Generalize

To investigate the effect of consolidation on explicit and implicit learners' structural transfer in Experiment 2 ($n=161$ after exclusions; see methods for details), we closely followed the design of Experiment 1 but introduced a 12-hour overnight consolidation phase between Phases 1 and 2. We found that while the performance of the explicit learners ($n=21$) after overnight sleep consolidation did not change drastically, it completely altered the pattern of behaviour of implicit learners ($n=140$) as they demonstrated in this condition the same generalization as the explicit learners did in Experiment 1 (Figure 2).

Specifically, explicit learners ($n=21$) performed above chance for pairs of the same structure ($M=66.7$, $SE=5.8$, $d=0.63$, $t(20)=2.87$ $p=0.038$, $BF=5.3$) but not pairs of a novel structure ($M=51.2$, $SE=5.8$, $d=0.04$, $t(20)=0.20$ $p=0.841$, $BF=0.23$) in the test of Phase 2 pairs. Implicit learners ($n=140$) also performed above chance for pairs of the same structure ($M=58.8$, $SE=2.5$, $d=0.30$, $t(139)=3.51$ $p=0.004$, $BF=37.6$), but not for pairs of the novel structure ($M=46.8$, $SE=2.6$, $d=0.11$, $t(139)=-1.24$ $p=0.435$, $BF=0.27$). The performance difference between these two types of pairs was significant ($M_{diff}=11.96$, $d=0.24$, $t(139)=2.82$ $p=0.027$, $BF=5.4$). A direct comparison of implicit participants' performance in Experiments 1 and 2 showed that the participants performed higher after sleep

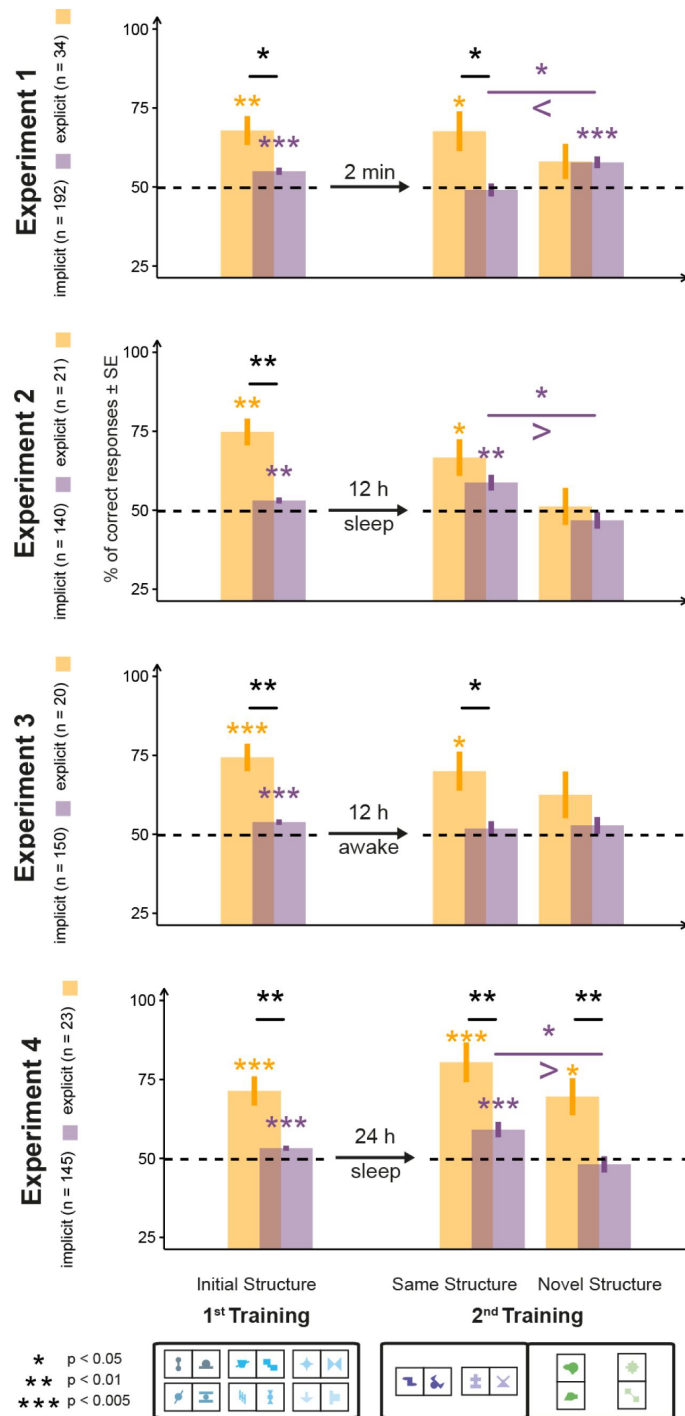


Figure 2

Results of familiarity tests in Experiments 1, 2, 3, and 4. Test results of the 2AFC trials in all four experiments are grouped on the x-axis according to whether the trials used shapes of the 1st or the 2nd training and, within the 2nd training, whether the pair in the trial had the same or different orientation structure as inventory pairs in the 1st training. The y-axis represents the proportion of correct responses in the 2AFC test trials. Arrows and text between the test results related to the two trainings convey the condition and length of the break period. Bars represent SEM, color coding indicates implicit and explicit subgroups of the participants. The horizontal dotted line denotes chance performance. Asterisks above bars denote significance levels from chance, while above lines, significance level comparing two conditions below the tips of the line. Legend of significance levels is shown in the lower left corner. Signs of inequality below the comparison in the 2nd training indicate the direction of effect.

for same structure pairs and lower for novel structure pairs (see next sections for detailed results). Overall, the results of Experiments 1 and 2 demonstrate that while explicit participants could immediately generalize the structure they learned, implicit participants required a consolidation period before being able to do the same.

The Effect of Consolidation is Specific to Sleep

To clarify whether the effect observed in Experiment 2 is specific to sleep or just a general effect of consolidation, in Experiment 3 ($n=170$ after exclusions; see methods for details), we used the same general procedure as in Experiment 2 but with the 12-hour consolidation phase occurring during the day. The results of the implicit learners ($n=150$) provided strong evidence against generalization, as they performed at chance with both pairs of the same structure ($M=51.8$, $SE=2.3$, $d=0.07$, $t(149)=0.80$ $p=0.999$, $BF=0.17$) and pairs of a novel structure ($M=52.8$, $SE=2.6$, $d=0.09$, $t(149)=1.08$ $p=0.999$, $BF=0.22$) of Phase 2 (**Figure 2**). In contrast, participants with explicit knowledge ($n=20$) replicated the results of Experiments 1 and 2, with percent correct above chance for pairs of the same structure ($M=70.0$, $SE=6.2$, $d=0.72$, $t(19)=3.24$ $p=0.026$, $BF=10.4$) but not for pairs of a novel structure ($M=62.5$, $SE=7.4$, $d=0.38$, $t(19)=1.70$ $p=0.530$, $BF=0.78$).

The Effect of Sleep is not Explained by a Time-of-Day Effect

It has previously been suggested that an apparent effect of sleep shown in an AM-PM vs PM-AM design can be based on the time of day at testing rather than sleep itself (38). To control for this potential confound, Experiment 4 ($n=169$ after exclusions; see methods for details) replicated Experiment 2 but with the second session delayed by 24 instead of 12 hours. Participants in this condition, therefore, had overnight sleep but were tested at the same time of day as the non-sleeping participants in Experiment 3. The results of Experiment 4 replicated the results of Experiment 2 for implicit learners, showing stronger learning for pairs of the same structure as compared to pairs of the novel structure ($M_{diff}=11.04$, $d=0.23$, $t(144)=2.75$ $p=0.021$, $BF=4.4$). We, therefore, conclude that the difference for implicit learners between our previous sleep and non-sleep conditions was not based on a time-of-day effect. For explicit learners, Experiment 4 showed no significant difference between learning for pairs of the same structure as compared to pairs of the novel structure ($M_{diff}=10.78$, $d=0.26$, $t(22)=1.27$ $p=0.437$, $BF=0.44$) and no significant correlation between learning pairs of the first learning phase and pairs of the novel structure ($r=0.109$, $p=0.620$). This suggests that the generalization effect found for explicit learners in the previous experiments is weakened during the longer consolidation phase of Experiment 4.

To compare directly the differential effect of type of consolidation on implicit structure learning, we entered the data of participants with implicit knowledge from Experiments 1, 2, 3, and 4 into a 4×2 ANOVA, with consolidation type (no consolidation, 12-h-sleep, 12-h-awake, and 24-h-sleep consolidation) and pair type (same or novel structure) as factors. The obtained results showed the typical pattern of a cross-over interaction with no significant main effects (consolidation type: $F(3,623)=0.18$, $p=0.910$, $BF=0.003$, $\eta_p^2=0.0009$; pair type: $F(1,623)=1.52$, $p=0.218$, $BF=0.17$, $\eta_p^2=0.002$) but a significant interaction ($F(3,623)=7.43$, $p<0.001$, $BF=1979$, $\eta_p^2=0.03$). Post-hoc tests revealed significant differences between the no-consolidation group (Exp. 1) and the two asleep-consolidation groups (Exp. 2 and 4), where the no-consolidation group showed stronger learning of novel structure pairs (Exp. 1 vs. Exp. 2: $p=0.004$, $BF=44.3$; Exp. 1 vs. Exp. 4: $p=0.012$; $BF=12.7$), while the asleep-consolidation groups showed stronger learning of same structure pairs (Exp. 1 vs. Exp. 2: $p=0.015$, $BF=8.8$; Exp. 1 vs. Exp. 4: $p=0.011$; $BF=14.3$). No other significant differences were found. These results confirm for implicit learners a directly opposite pattern of generalization behaviour between no-consolidation and asleep-consolidation conditions.

Additionally, to test for the presence of potential time-of-day effects in Experiment 1, we reanalyzed the data of Experiment 1 by taking into account the time point of testing. Both correlational and subgroup analyses found no indication of an effect of time of day on the pattern of structural transfer (see Supplementary Material).

The Type of Transfer Depends on Quality of Knowledge, Not Quantity of Knowledge

The different patterns of structural transfer for explicit and implicit participants could be based on either the quality of knowledge, i.e., its explicitness, or the quantity of knowledge, i.e., how much was learned during the first training phase. However, these two factors are confounded since the explicit participants typically performed higher for pairs of the first training phase. To address this confound, we conducted a matched sample analysis⁽³⁹⁾ in Experiments 1, 2, 3, and 4 to clarify which aspect of knowledge was responsible for our results. We selected a subsample of our implicit participants so that their accuracy performance matched that of the explicit participants for the first training phase and performed the same analyses on these subsampled populations as on the entire dataset in Experiments 1, 2, 3, and 4 (for details and a discussion of potential shortcomings see Supplementary Materials). Matched implicit participants showed the same overall pattern of generalization behaviour as the full sample of implicit participants for all four experiments (**Figure 3**), although this failed to reach significance for Experiment 4. Specifically, participants learned more pairs of the novel than of the same structure in Experiment 1 ($p=0.012$; $BF=3.6$), they learned more pairs of the same than of the novel structure in Experiment 2 ($p<0.001$, $BF=127$), and they showed no significant difference between learning the two types of pairs in Experiment 3 ($p=0.214$, $BF=0.59$) and Experiment 4 ($p=0.304$, $BF=0.46$). With the matched sample analysis, we drastically reduced the sample size of the implicit participants, therefore reducing the power to detect the small effects found for this group. However, we recovered the same descriptive pattern in the full data set and the matched sample groups in all experiments and failed to recover the same statistical significance only in the time-of-the-day control experiment. Therefore, we posit that these findings strongly support the notion that the difference in the structural transfer is based predominantly on the quality of knowledge - its explicitness - not the exact level of learning.

Explicitness and Generalization can be Induced by Verbal Instruction

The explicit-implicit parameterization in Experiment 1 was quasi-experimental rather than true experimental since the groups were formed naturally. Therefore, it is unclear whether it is indeed explicitness that enables generalization or whether the two groups of participants are different in other important ways (e.g., task engagement, and attentional processes). To investigate this issue, in Experiment 5 ($n=36$ after exclusions), we rerun Experiment 1 with a new group of participants and with the single change in the protocol that explicitness of knowledge was induced via explicit verbal instructions stating that all shapes would be grouped into pairs. The results of Experiment 5 closely matched those of the explicit participants of Experiment 1. The participants had an above chance performance for Phase 1 ($M=71.8$, $SE=3.6$, $d=1.01$, $t(35)=6.07$ $p<0.001$, $BF=26395$) and for the same structure ($M=72.9$, $SE=4.0$, $d=0.95$, $t(35)=5.69$ $p<0.001$, $BF=9058$) but not novel structure pairs ($M=52.8$, $SE=5.8$, $d=0.08$, $t(35)=0.48$ $p=0.634$, $BF=0.20$) in Phase 2. The performance for pairs of the same and novel structure was significantly different ($d=0.64$, $t(35)=3.83$ $p=0.001$, $BF=58.0$). These results confirm that the generalization producing an immediate transfer between contexts can be easily induced by verbal instructions in any population and therefore, the type of explicitness studied in our experiments is a likely candidate of the necessary condition for such a generalization.

In Experiments 1-4 and Supplementary Experiment 1, explicit participants showed the same descriptive pattern of higher performance for same over novel structure pairs, but this difference failed to reach significance within experiments. Therefore, we reanalyzed these results by collapsing the data over all these experiments. The results showed an overall higher performance on same structure pairs than novel structure pairs for explicit participants ($d=0.34$, $t(233.99)=2.56$,

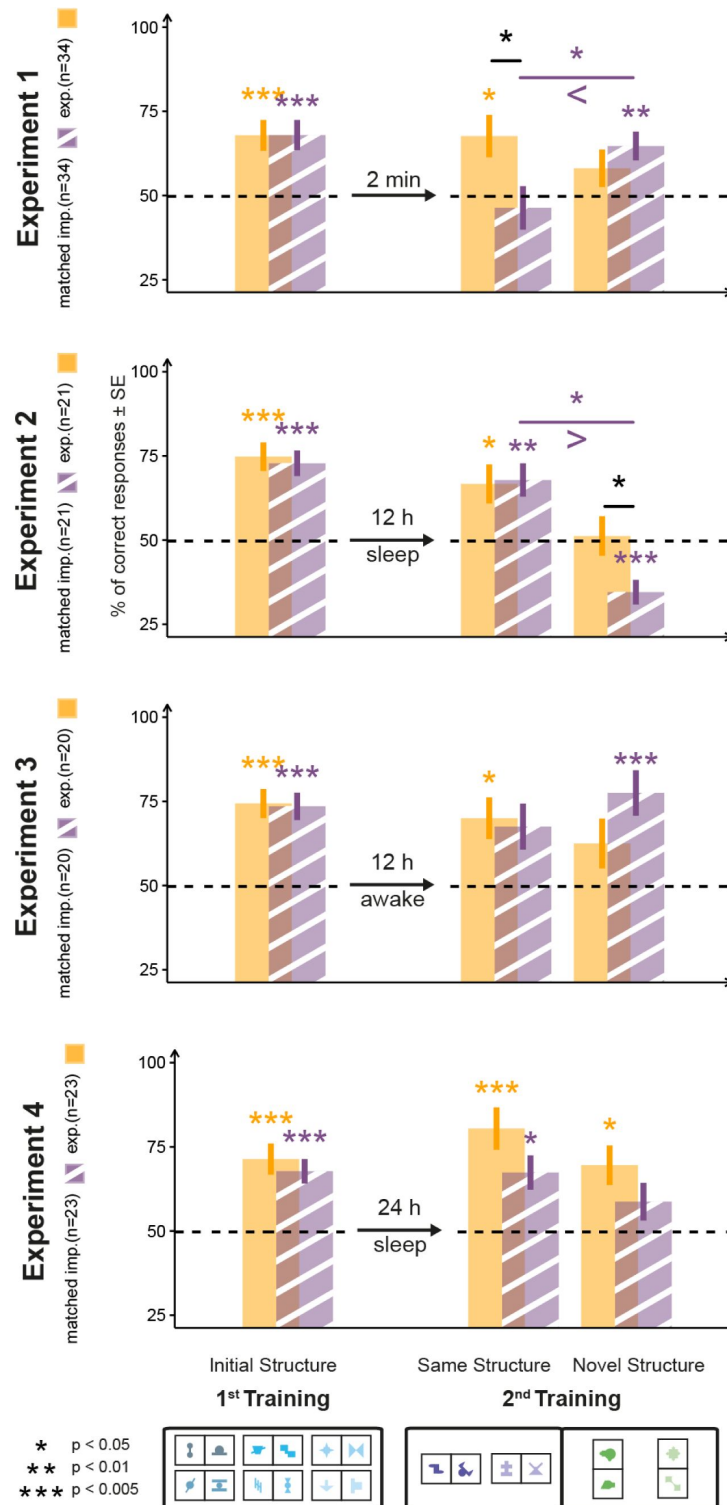


Figure 3.

Matched Sample Analysis of Experiments 1, 2, 3, and 4. The structure of the figure is identical to that of **Figures 1** and **2**, with data for the explicit participants (orange bars) being the same as in **Figure 1**, while data for the implicit participants (striped purple bars) showing the subgroup of implicit participants from Experiment 1 whose combined performance matched the performance of the explicit participants on test trials of the first learning phase. The y-axes represent the proportion of correct responses in the 2AFC trials. Bars represent the mean ($\pm$ SEM) for each type of pair (pairs of Phase 1 and same and novel structure pairs of Phase 2). The horizontal dashed line indicates chance performance.

$p=0.011$, $BF=3.12$). Furthermore, over all experiments, explicit participants' performance for Phase 1 only significantly correlated with learning of same structure pairs ($r=0.41$, $p<0.001$, $BF=4460$), not novel structure pairs ($r=0.14$, $p=0.137$, $BF=0.62$).

Discussion

Our findings demonstrated for the first time an interaction of consolidation and explicitness of knowledge in humans' unsupervised transfer learning. Participants with explicit knowledge are able to immediately transfer structural knowledge from one learning context to another. In contrast, participants with implicit knowledge show a structural novelty effect in immediate transfer. Only after a phase of sleep consolidation are they able to generalize from one learning context to another.

Our interpretation of these results is that sleep consolidation leads to an internal redescription of the acquired knowledge by factorization, that is, by representing the feature of orientation as a useful summary statistic in its own right. This redescription of the input with a hierarchical model using a higher-level representation of the underlying orientation structure emerges as a complement to the chunk representation learned by statistical learning during exposure. While without such a redescription, the previously learned patterns in Phase 1 interfere with new ones in Phase 2. When redescription occurs, the abstract knowledge at the higher level of the hierarchy can be generalized i.e., it can be used during the processing and segmentation of new input. Instead of a simple feed-forward hierarchy progressing from specific to abstract, our interpretation implies a hierarchical system with bi-directional interactions: learning some representations at a higher level of abstraction (orientation) constrains the learning of new representations of lower-level features (actual chunks) later in time. Thus our approach to hierarchy is different from earlier treatments of SL in hierarchical systems that focused on hierarchies of composition (36) or hierarchies that arbitrated between competing learning systems (40).

One notable consequence of the emergence of such hierarchies is linking SL more to the emergence of object representations. Previous studies demonstrated that statistical learning is not based solely on momentarily observable co-occurrence statistics but is also biased by prior knowledge. However, these studies demonstrate various effects of prior knowledge acquired over a lifetime (41) or shorter period (42–44), provided explicitly (45), or facilitated by segmentation cues (46). They also focused mostly on the language domain or sequential streams and either relied on an uncontrolled amount of long-term knowledge or investigated token-level effects. In contrast, the current study investigated structure-related effects that could not be explained by the transfer of token-level knowledge, and it strictly controlled the amount of previous exposure to the target structure. Our results demonstrate that similar to objects (47, 48), chunks acquired during SL are not represented simply as an unrelated set of inventory elements but organized into higher-level categories based on the common underlying structures - e.g., the category of horizontal pairs. Thus, our results build a bridge between classical SL and category learning, and, in line with earlier findings (49, 50), it supports the idea that SL is a vehicle for acquiring object-like representations.

Previous studies on statistical learning and consolidation focused on how much time or sleep helps with the stabilization or improvement of specific memory traces and obtained mixed results, with some studies showing clear effects (20, 51–53) and others showing no or very limited effects of sleep (54–60). Our results are in line with the latter set of studies, showing no evidence of a direct effect of sleep on statistical learning per se; i.e. no effect on the learning of specific item-item associations. In addition, most of these studies did not consider the actual state of the acquired knowledge (but see (29)). In contrast, our study specifically focused on the difference between explicit and implicit learners to disentangle the complex nature of

consolidation and we use the ability to generalize structure between learning contexts as our primary measure. This approach allowed us to compare our findings to those reported in the domain of explicit learning (20, 21, 24, 61), under names such as schemas (20), rules (24), insight (26), and gist (62), and uncover a surprising similarity between consolidation benefits based on abstraction and generalization in the two domains. In particular, similarly to observers who needed sleep to “discover” abstract learning rules in an explicit task (61), our implicit learners required sleep to “uncover” abstract higher-level descriptors of the observed structures and utilize them in subsequent implicit learning. This parallel in behaviour suggests an analogous process for integrating and using implicitly and explicitly acquired new information at different levels of the existing internal representation.

A potential shortcoming of the current study is that, for practical reasons and due to the online nature of the experiments, the assignment of participants to the different consolidation conditions was not fully randomized. Instead, the various studies were made available at different points in time (i.e., data collection for Experiment 2 commenced after data collection for Experiment 1 was finished, etc.). This could, in principle, lead to a sampling bias due to preferences for conditions. Although we don’t believe that it is likely that such a sampling bias on its own could produce the observed results in lieu of full experimental control, we cannot rule it out entirely.

Our results and interpretations for Experiment 1 suggest that for implicit participants learning in Phase 2 was subject to structure-specific proactive interference in the absence of consolidation as learning of same-structure pairs was hindered by the previous learning of pairs in Phase 1. Note that these findings do not exclude the possibility of simultaneously active retroactive interference. To establish if such an effect was present in the current study would require a further control condition of participants receiving only the Phase 1 training. We did not investigate this condition since the effect of retroactive interference was not the focus of the current study, and any obtained results would not alter our conclusions.

Previous statistical learning experiments reported that quantitative performance improves by explicit instructions (45) and that performances in the presence and absence of explicit instructions converge over longer exposure (63). Our results in Experiment 5 are in line with those findings by showing a stronger performance after explicit instructions about the task structure and a convergence between the outcomes of explicit and implicit learning. However, we also go beyond these findings by demonstrating that vanishing of differences after sleep consolidation occurs even when the difference between the initial implicit and explicit learning results prior to consolidation is qualitative rather than only quantitative.

In summary, our work builds multiple bridges between different domains of cognition. First, it relates low-level correlational learning and higher-level structure learning in an unsupervised transfer learning setup that is separate from the classical problem- or task-solving paradigms (26, 64, 65) yet can be readily interpreted in those frameworks. This extends the definition of higher-level structure to a domain where explicit tasks and thoughts are not defined. Second, it reports a new repulsion-attraction effect at the level of implicit structure transfer, namely a tradeoff between generalization and interference depending on the presence or absence of sleep-consolidation processes. Previously, such repulsion-attraction effects have been found either within immediate perceptual processes such as implicit low-level perception (66–68) but without linking it to sleep or with high-level categorization (20, 21, 24, 61, 69) that involved sleep but also used explicit tasks. These converging findings might indicate the existence of a more general factorization mechanism in the brain that works continuously and at multiple levels during perceptual and cognitive processes following the same computational principles. However, confirming this conjecture will require further investigations of the interplay between learning and consolidation in humans. Our approach could help to resolve this issue by facilitating an integrated exploration of how the brain learns complex internal representations implicitly and utilizes them in a recursive manner.

Data Availability

The experimental data that support the findings of this study are available on OSF: <https://osf.io/untbz> 

Methods

Experiment 1

Participants

251 participants (92 female, Age: mean = 28.0, mode = 25, SD = 9.5) were recruited via [prolific.co](#). All participants had normal or corrected-to-normal vision and provided informed consent. The study was approved by the Hungarian United Ethical Review Committee for Research in Psychology (EPKEB) and all participants provided informed consent. The hourly compensation was £ 6.3. The sample size was chosen to achieve 80% power for expected small effect sizes ($d=0.2$) in paired t-tests (needed sample for the implicit subgroup according to power analysis = 198.15) and to account for exclusions.

Materials

The stimuli were taken from [\(31\)](#) and consisted of 20 abstract black shapes on white background (see **Figure 1** ). The shapes were grouped to form six pairs of the same orientation (horizontal or vertical) for the first learning phase and four pairs, two horizontal and two vertical, for the second learning phase. The assignment of shapes to pairs was randomized for each participant. Scenes were created by placing three pairs together on a 3×3 grid without segmentation cues. 160 scenes were created for the first and 48 for the second learning phase. In the second learning phase, each scene was used twice for a total of 96 presented scenes.

The study was conducted online via [prolific](#). As this was an online study, participants conducted it on their own computers using google chrome, safari, or opera browser. Only desktop and laptop computers were admissible, and no smartphones or tablets. Stimuli were presented using custom JavaScript code built on the jsPsych library [\(70\)](#). As participants used different devices (screen size and resolution) the visual angle of the shapes was not the exact same for all participants. Instead, the 3×3 grid extended over 600×600 pixels and was centered in the middle of the screen. The remaining screen outside the grid was empty (white).

Procedure

Participants passively observed 160 scenes in the first training phase. For half of the participants, these scenes contained only horizontal pairs (horizontal condition), and for the other half, only vertical pairs (vertical condition). Each scene was presented for 2 s with a 1 s interstimulus interval (ISI). After a two-minute passive break, participants passively observed 96 scenes in the second training phase. Participants were not told about the presence of any structure in the scenes and were simply instructed to be attentive so that they could later answer simple questions. After half of each training phase, an attention check appears, asking participants to press the spacebar to continue. Response time for the attention check was recorded to detect inattentive participants. After the second training phase, participants had another two-minute passive break.

Following this, pair learning was tested with a two-alternative forced choice task (2AFC). In each trial, participants saw a real shape pair from one of the training phases and a foil pair created by combining shapes from two different pairs of the same training phase. Real and foil pairs were

presented after each other in the 3×3 grid for 2 s with a 1 s ISI. The order of real and foil pairs was randomized. Participants were asked to indicate which of the two was more familiar by pressing “1” or “2” on their keyboard. Participants first completed 16 trials using pairs from the second training phase, where each of the 4 real pairs of that training phase was tested against 2 different foil pairs twice. This was followed by 24 trials using pairs from the first training phase, where each of the 6 real pairs of that training phase was tested against 4 different foil pairs. Overall, all real and foil pairs were used the same number of times during the test phase. Finally, participants answered five open questions about their beliefs about the experiment and their knowledge of pair structure (see Supplementary Materials for details).

Results

Based on pilot data we choose 20 seconds combined response time for both attention checks as the cut-off value for inclusion. 19 participants were rejected for failing this criterion. Response bias was defined as the proportion with which participants used one of the two response options (“1” and “2”) and participants who were 2.5 SD away from the mean were excluded. 3 participants were excluded for failing this criterion. This left us with 229 participants after exclusions. Based on the open responses at the end of the experiment participants were categorized into one of three groups. Participants who reported no knowledge of pairs were counted as implicit ($n=192$), participants who reported knowledge of the presence of pairs were counted as explicit ($n=34$), and participants who also reported the underlying horizontal/vertical structure were excluded from analysis as they were too few for meaningful analysis ($n=3$). For details on the exit survey and example responses see Supplementary Materials.

Test trials for the second training phase were scored separately for pairs of the same structure as in the first training phase and pairs of the novel structure. Explorative analysis of the trials of the second training phase revealed a strong negative correlation between trials of the novel and the same structure ($r=-0.432$, $p<0.001$). As the foil pairs used for one type of structure were created by recombining shapes of pairs of the other structure, this could be a type of consistency effect where participants tend to choose the same shapes, independent of pair knowledge. Specifically, we assume that in addition to applying the weak pair knowledge acquired during the training phase, participants also consider the specific shapes seen in a test trial for their response. I.e., after a participant decided against a specific foil pair in an early test trial, they are less likely to pick an option using these same shapes in later test trials. Due to the foil pairs always being constructed of only shapes of pairs of the opposite structure, this would lead to participants developing a bias for only one type of structure (i.e., one set of shapes) in the course of the testing phase. If this is the case we expect to see an increase in this negative correlation as participants complete more test trials. Indeed, for the first 8 trials, this correlation is $r=-0.209$, while it increases to $r=-0.435$ in the second 8 trials. Based on this exploration we included only the first 8 trials in all the following analyses to minimize this consistency effect. To keep comparability, the following experiments use the same number of test trials but we again only analyze the first 8.

The data was collapsed over vertical and horizontal conditions for all further analysis, as a 3×2 mixed ANOVA with *test type* (levels: training 1, same structure, novel structure) as within-subject factor and *condition* (levels: horizontal, vertical) as between-subject factor showed no significant main effect of condition ($F(1, 224)=0.776$, $p=0.379$, $BF=0.08$, $\eta_p^2=0.003$) and no significant *test type* - *condition* interaction ($F(2,448)=0.199$, $p=0.820$, $BF=0.04$, $\eta_p^2=0.001$).

Bayes Factors (BF) reported for the ANOVA here and throughout the text are based on Bayesian ANOVAs using the *BayesFactor* R package, realizing Bayesian tests with models analogous to the frequentist counterpart, and employing a JZS prior (71). BFs reported for t-tests throughout the text are calculated with the same package, following the same logic, and again employing the JZS prior unless specified otherwise. All other settings for the R functions of the *BayesFactor* package

where left at the default values, unless specified otherwise. P-values reported throughout the text were subject to experiment-wise correction for multiple comparisons using the Holm-Bonferroni method. All conducted tests are two-sided, unless specified otherwise.

The results (**Figure 2**) show above chance performance for the first training phase for both explicit ($M=67.9$, $SE=4.6$, $d=0.67$, $t(33)=3.93$ $p=0.002$, $BF=70.8$) and implicit ($M=55.0$, $SE=1.1$, $d=0.32$, $t(191)=4.46$ $p<0.001$, $BF=919$) participants. However, the performance was significantly higher for explicit participants ($d=0.73$, $t(224)=2.75$ $p=0.009$), as shown by a Welch's t-test assuming unequal variances (this result does not contain a Bayes Factor as there is no readily available implementation for Bayes Factor calculation for Welch's t-test). For the second learning phase, implicit participants show above chance performance for pairs of the novel structure ($M=57.8$, $SE=1.9$, $d=0.30$, $t(191)=4.16$ $p<0.001$, $BF=281$), and chance performance for pairs of the same structure as before ($M=49.1$, $SE=2.1$, $d=0.03$, $t(191)=-0.44$ $p=0.659$, $BF=0.09$). The performance for these types of pairs is significantly different from each other ($d=0.20$, $t(382)=2.78$ $p=0.030$, $BF=3.38$). Explicit participants show the opposite pattern with above-chance performance for pairs of the same structure as before ($M=67.6$, $SE=6.3$, $d=0.48$, $t(33)=2.81$ $p=0.033$, $BF=5.0$), and chance performance for pairs of the novel structure ($M=58.1$, $SE=5.6$, $d=0.25$, $t(33)=1.46$ $p=0.465$, $BF=0.48$). However, for them, this difference is not significant ($d=0.19$, $t(33)=1.12$ $p=0.539$, $BF=0.33$). However, we see a significant, medium to large correlation between learning in the first training phase and learning pairs of the same structure in the second training phase ($r=0.45$, $p=0.008$) for the explicit participants. Such a correlation is not observed between learning in the first learning phase and learning pairs of the novel structure in the second learning phase ($r=-0.01$, $p=0.947$). For implicit participants, neither of the two correlations are significant (*same structure*: $r=0.01$, $p=0.853$; *novel structure*: $r=0.04$, $p=0.597$); likely because the small effect sizes away from chance for this group lead to high noise in the data.

The different patterns of transfer behaviour for explicit and implicit participants are also confirmed when entering the data into a 2×2 mixed ANOVA with factors *participant type* (explicit or implicit) and *structure type* (novel or same). The results show a significant main effect of *participant type* ($F(1, 224)=8.03$, $p=0.005$, $BF=2.1$, $\eta_p^2=0.03$) and of *structure type* ($F(1, 224)=4.05$, $p=0.045$, $BF=1.2$, $\eta_p^2=0.02$) as well as a significant interaction of both ($F(1,224)=4.89$, $p=0.028$, $BF=31.6$, $\eta_p^2=0.02$). Post-hoc analysis of achieved power in *gpower* (72) using the values for the interaction effect and the correlation among the repeated measures reported above suggested that the achieved power for this test was 0.79.

To test for a possible time-of-day effect in learning or generalization we correlated test performance with the hour of the day at which participants completed the experiment. There were no significant correlations for pairs of the same structure (explicit participants: $r=-0.03$, $p=0.882$; implicit participants: $r=0.01$, $p=0.893$) or pairs of the novel structure (explicit participants: $r=0.13$, $p=0.477$; implicit participants: $r=-0.05$, $p=0.458$). Additionally, we looked separately at groups of participants completing the experiment early in the day (7-11 am) and late in the day (7-11 pm). For implicit participants, there was no significant difference between participants that participated early ($n=23$) or late ($n=22$) as a 2×2 mixed ANOVA with *hour-of-day* and *test type* as factors showed no significant main effect of *hour-of-day* ($F(1,43)=0.019$, $p=0.892$, $BF=0.26$) and no significant *hour-of-day* – *test type* interaction ($F(1,43)=0.095$, $p=0.759$, $BF=0.31$).

Experiment 2

Participants

259 participants (127 female, Age: mean = 25.6, mode = 21, SD = 8.5) were recruited via prolific.co. All participants had normal or corrected-to-normal vision and provided informed consent. The sample size was chosen to match that of Experiment 1. The study was approved by the Psychological Research Ethics Board of the Central European University and all participants

provided informed consent. The hourly compensation was £ 6.3. To ensure that participants have overnight sleep during the experiment as intended several constraints and checks were implemented (see *Sleep During Consolidation Studies* section of the supplementary material).

Materials

The materials of the main part of the experiment were identical to Experiment 1. Additionally, participants filled out the Pittsburgh Sleep Quality Index (PSQI)([73](#)) and the Groningen Sleep Quality Scale (GSQS)([74](#)).

Procedure

The procedure within the main tasks was identical to Experiment 1. However, in this experiment, participants completed the first training phase in the evening at 09:00 pm, followed by the GSQS and PSQI questionnaires. 12 hours later in the morning at 09:00 am, they completed the second training phase, followed by all the test trials and finally, they completed the GSQS again.

Results

The same exclusion criteria as Experiment 1 were used. This led to 18 exclusions for failed attention checks and 2 exclusions for response bias. In addition to the exclusion criteria used in Experiment 1, here we also employed exclusion criteria related to sleep quality. 47 participants were excluded because they reported bad sleep quality for the night before the experiment or the night of the experiment as measured by the Groningen Sleep Quality Scale (GSQS, score below 9). Additionally, 31 participants were excluded because they reported a bad habitual sleep quality measured with the Pittsburgh Sleep Quality Index (PSQI, score below 10). This left us with 161 participants after exclusions. Participants were categorized as explicit or implicit in the same way as in Experiment 1. This led to 21 participants being categorized as explicit and 140 as implicit.

Bayes Factors from Bayesian t-tests for implicit participants reported for Experiments 2, 3, 4, and Supplementary Experiment 1 used an r -scale parameter of 0.5 instead of the default $\sqrt{2}/2$. The r -scale parameter controls the prior over effect sizes and using 0.5 means increasing the prior probability of small effects. This decision was made to incorporate the shifted belief about expected effect sizes for implicit participants after observing the effects in Experiment 1.

The results ([Figure 2](#)) showed that participants with implicit knowledge ($n=140$) performed above chance for pairs of the first training phase ($M=53.1$, $SE=1.0$, $d=0.27$, $t(139)=3.24$ $p=0.009$, $BF=16.8$) and for pairs of the same structure ($M=58.8$, $SE=2.5$, $d=0.30$, $t(139)=3.51$ $p=0.004$, $BF=37.6$) but not pairs of a novel structure ($M=46.8$, $SE=2.6$, $d=0.11$, $t(139)=-1.24$ $p=0.435$, $BF=0.27$) in the second training phase. The performance for pairs of the same and novel structure is significantly different ($d=0.24$, $t(139)=2.82$ $p=0.027$, $BF=5.4$). Participants with explicit knowledge ($n=21$) show the same pattern of results as they did in Experiment 1, performing above chance for pairs of the first training phase ($M=74.8$, $SE=4.2$, $d=1.3$, $t(20)=5.9$ $p<0.001$, $BF=2361$) and for pairs of the same structure ($M=66.7$, $SE=5.8$, $d=0.63$, $t(20)=2.87$ $p=0.038$, $BF=5.3$) but not pairs of a novel structure ($M=51.2$, $SE=5.8$, $d=0.04$, $t(20)=0.20$ $p=0.841$, $BF=0.23$) in the second training phase. As in Experiment 1, for explicit participants, the difference between pairs of the same and novel structure is not significant ($d=0.39$, $t(20)=1.77$ $p=0.272$, $BF=0.86$), but we do see a strong positive correlation between learning in the first learning phase and learning pairs of the same structure in the second learning phase ($r=0.52$, $p=0.015$). Such a correlation is again not observed between learning in the first learning phase and learning pairs of the novel structure in the second learning phase ($r=0.27$, $p=0.232$). For implicit participants neither of the two correlations are significant (*same structure*: $r=0.05$, $p=0.570$; *novel structure*: $r=-0.02$, $p=0.788$); likely because the small effect sizes away from chance for this group lead to high noise in the data.

Experiment 3

Participants

275 participants (134 female, Age: mean = 28.9, mode = 24, SD = 9.8) were recruited via prolific.co. All participants had normal or corrected-to-normal vision and provided informed consent. The sample size was chosen to match that of Experiment 1. The study was approved by the Psychological Research Ethics Board of the Central European University and all participants provided informed consent. The hourly compensation was £ 6.3. To ensure that participants have overnight sleep during the experiment as intended several constraints and checks were implemented (see *Sleep During Consolidation Studies* section of the supplementary material).

Materials

The materials were identical to those of Experiment 2.

Procedure

The procedure was the same as Experiment 2, with the difference that the first session took place in the morning at 09:00 am and the second in the evening at 09:00 pm. As there was no night of sleep between the first and second sessions, participants filled out the GSQS only once in this experiment.

Results

The same exclusion criteria as Experiment 2 were used. This led to 29 exclusions for failed attention checks and 3 exclusions for response bias. 28 participants were excluded because they reported bad sleep quality for the night before the experiment as measured by the Groningen Sleep Quality Scale (GSQS, score below 9). 28 participants were excluded because they reported a bad habitual sleep quality measured with the Pittsburgh Sleep Quality Index (PSQI, score below 10). Additionally, 17 participants in this experiment were excluded as they reported that they slept at all during the day. This left us with 170 participants after exclusions. Participants were categorized as explicit or implicit in the same way as in Experiment 1. This led to 20 participants being categorized as explicit and 150 as implicit.

The results (**Figure 2** [↗](#)) showed that participants with implicit knowledge ($n=150$) that did not sleep during the consolidation phase performed above chance for pairs of the first training phase ($M=53.9$, $SE=0.9$, $d=0.36$, $t(149)=4.46$ $p<0.001$, $BF=1020$), but not for pairs of the same structure ($M=51.8$, $SE=2.3$, $d=0.07$, $t(149)=0.80$ $p=0.999$, $BF=0.17$) or pairs of a novel structure ($M=52.8$, $SE=2.6$, $d=0.09$, $t(149)=1.08$ $p=0.999$, $BF=0.22$) in the second training phase. Participants with explicit knowledge ($n=20$) again perform as they did in Experiment 1 performing above chance for pairs of the first training phase ($M=74.4$, $SE=4.3$, $d=1.27$, $t(19)=5.67$ $p<0.001$, $BF=1281$) and for pairs of the same structure ($M=70.0$, $SE=6.2$, $d=0.72$, $t(19)=3.24$ $p=0.026$, $BF=10.4$) but not pairs of a novel structure ($M=62.5$, $SE=7.4$, $d=0.38$, $t(19)=1.70$ $p=0.530$, $BF=0.78$) in the second training phase. As in Experiment 1, for explicit participants, the difference between pairs of the same and novel structure is not significant ($d=0.19$, $t(19)=0.86$, $p=0.999$, $BF=0.32$), but again we do see a strong positive correlation between learning in the first learning phase and learning pairs of the same structure in the second learning phase ($r=0.56$, $p=0.010$). Such a correlation is again not observed between learning in the first learning phase and learning pairs of the novel structure in the second learning phase ($r=0.42$, $p=0.065$). For implicit participants, the *same structure pairs* ($r=0.07$, $p=0.430$) correlation is not significant, while we see a small significant correlation for the *novel structure pairs* ($r=0.18$, $p=0.025$).

Experiment 4

Participants

275 participants (129 female, Age: mean = 27.9, mode = 23, SD = 8.9) were recruited via [prolific.co](#). All participants had normal or corrected-to-normal vision and provided informed consent. The sample size was chosen to match that of Experiment 1. The study was approved by the Psychological Research Ethics Board of the Central European University and all participants provided informed consent. The hourly compensation was £ 6.3. To ensure that participants have overnight sleep during the experiment as intended several constraints and checks were implemented (see *Sleep During Consolidation Studies* section of the supplementary material).

Materials

The materials were identical to Experiment 2.

Procedure

The procedure was identical to Experiment 2. However, in this experiment participants completed the first session in the evening at 09:00 pm, and the second session 24 hours later again in the evening at 09:00 pm.

Results

The same exclusion criteria as Experiment 2 were used. This led to 17 exclusions for failed attention checks, and 5 exclusions for response bias. 51 participants were excluded because they reported bad sleep quality for the night before the experiment or the night of the experiment as measured by the Groningen Sleep Quality Scale (GSQS, score below 9). Additionally, 33 participants were excluded because they reported a bad habitual sleep quality measured with the Pittsburgh Sleep Quality Index (PSQI, score below 10). This left us with 169 participants after exclusions. Participants were categorized as explicit or implicit in the same way as in Experiment 1. This led to 145 participants being categorized as implicit, 23 categorized as explicit, and 1 being excluded for being categorized as fully explicit.

The results (**Figure 2** [↗](#)) showed that participants with implicit knowledge ($n=145$) performed above chance for pairs of the first training phase ($M=53.2$, $SE=0.8$, $d=0.34$, $t(144)=4.09$ $p=0.001$, $BF=260$) and for pairs of the same structure ($M=59.1$, $SE=2.4$, $d=0.31$, $t(144)=3.78$ $p=0.001$, $BF=89.4$) but not pairs of a novel structure ($M=48.1$, $SE=2.6$, $d=0.06$, $t(144)=-0.74$ $p=0.459$, $BF=0.17$) in the second training phase. The performance for pairs of the same and novel structure is significantly different ($M_{diff}=11.04$, $d=0.23$, $t(144)=2.75$ $p=0.021$, $BF=4.4$). Participants with explicit knowledge ($n=23$) show the same pattern of results as they did in Experiment 1, performing above chance for pairs of the first training phase ($M=71.4$, $SE=4.6$, $d=0.9$, $t(22)=4.6$ $p=0.001$, $BF=217$), for pairs of the same structure ($M=80.4$, $SE=6.3$, $d=1.01$, $t(22)=4.85$ $p=0.001$, $BF=347$), and pairs of a novel structure ($M=69.6$, $SE=5.9$, $d=0.69$, $t(22)=3.33$ $p=0.012$, $BF=13.6$) in the second training phase. As in Experiment 1, for explicit participants, the difference between pairs of the same and novel structure is not significant ($M_{diff}=10.78$, $d=0.26$, $t(22)=1.27$ $p=0.437$, $BF=0.44$), but here we do not see a significant positive correlation between learning in the first learning phase and learning pairs of the same structure in the second learning phase ($r=0.109$, $p=0.620$) and no significant correlation between learning in the first learning phase and learning pairs of the novel structure in the second learning phase ($r=0.111$, $p=0.613$). For implicit participants, neither of the two correlations are significant (*same structure*: $r=-0.01$, $p=0.894$; *novel structure*: $r=0.01$, $p=0.940$); likely because the small effect sizes away from chance for this group lead to high noise in the data.

Experiment 5

Participants

40 participants (18 female, Age: mean = 28.4, mode = 19, SD = 11.8) were recruited via prolific.co. All participants had normal or corrected-to-normal vision and provided informed consent. The sample size was chosen to approximately match the number of explicit participants in Experiment 1 after exclusions. The study was approved by the Psychological Research Ethics Board of the Central European University and all participants provided informed consent. The hourly compensation was £ 6.3. To ensure that participants have overnight sleep during the experiment as intended several constraints and checks were implemented (see *Sleep During Consolidation Studies* section of the supplementary material).

Materials

The materials were identical to Experiment 1.

Procedure

The procedure was identical to Experiment 1, apart from the instructions. In this experiment, participants were told about the pair structure before the beginning of the experiment. Participants were told that whenever a specific shape appears in the grid a second specific shape will appear in a fixed position near it.

Results

The exclusion criteria were identical to those used in Experiment 1. This led to 4 exclusions for failing the attention checks, and 0 exclusions for response bias. This left us with 36 participants after exclusions. Overall, the results show the same pattern as the explicit participants in Experiment 1 (see **Supplementary Figure 1** [↗](#), top panel), with above-chance performance for pairs of the first training phase ($M=71.8$, $SE=3.6$, $d=1.01$, $t(35)=6.07$ $p<0.001$, $BF=26395$) and for pairs of the same structure ($M=72.9$, $SE=4.0$, $d=0.95$, $t(35)=5.69$ $p<0.001$, $BF=9058$) but not pairs of a novel structure ($M=52.8$, $SE=5.8$, $d=0.08$, $t(35)=0.48$ $p=0.634$, $BF=0.20$) in the second training phase. The performance for pairs of the same and novel structure is significantly different ($d=0.64$, $t(35)=3.83$ $p=0.001$, $BF=58.0$).

Analysis Across Experiments

To compare directly the differential effect of type of consolidation on implicit structure learning, we entered the data of participants with implicit knowledge from Experiments 1, 2, 3 and 4 into a 4×2 ANOVA, with *consolidation type* (no consolidation, 12-h-sleep, 12-h-awake, and 24-h-sleep consolidation) and *structure type* (same or novel structure) as factors. The obtained results showed the typical pattern of a cross-over interaction with no significant main effects (*consolidation type*: $F(3,623)=0.18$, $p=0.910$, $BF=0.003$, $\eta_p^2=0.0009$; *structure type*: $F(1,623)=1.52$, $p=0.218$, $BF=0.17$, $\eta_p^2=0.002$) but a significant interaction ($F(3,623)=7.43$, $p<0.001$, $BF=1979$, $\eta_p^2=0.03$). Post-hoc analysis of achieved power in *gpower* ([↗](#)) using the values for the interaction effect suggested that the achieved power was 0.99. Post-hoc tests revealed significant differences between the no-consolidation group (Exp. 1) and the two asleep-consolidation groups (Exp. 2 and 4), where the no-consolidation group showed stronger learning of novel structure pairs (Exp. 1 vs Exp. 2: $p=0.004$, $BF=44.3$; Exp. 1 vs Exp. 4: $p=0.012$; $BF=12.7$), while the asleep-consolidation groups showed stronger learning of same structure pairs (Exp. 1 vs Exp. 2: $p=0.015$, $BF=8.8$; Exp. 1 vs Exp. 4: $p=0.011$; $BF=14.3$).

In Experiments 1-4 and Supplementary Experiment 1 explicit participants showed the same descriptive pattern of higher performance for same over novel structure pairs, but this difference failed to reach significance within experiments. Therefore, we reanalyzed these results by collapsing the data over all these experiments. The results showed an overall higher performance on same structure pairs than novel structure pairs for explicit participants ($d=0.34$, $t(233.99)=2.56$, $p=0.011$, $BF=3.12$). Post-hoc analysis of achieved power in *gpower* (72 [↗](#)) effect suggested that the achieved power for this test was 0.96. Furthermore, over all experiments, explicit participants' performance for Phase 1 only significantly correlated with learning of same structure pairs ($r=0.41$, $p<0.001$, $BF=4460$), not novel structure pairs ($r=0.14$, $p=0.137$, $BF=0.62$). To validate the soundness of collapsing the data over experiments, we ran a 4×2 mixed ANOVA with factors Experiment and Test Type (Phase1, Novel Structure, Same Structure). The results showed the expected significant main effect of Test Type ($F(2,323)=5.52$, $p=0.004$, $BF=13.9$, $\eta_p^2=0.03$) and a main effect of Experiment ($F(4,323)=3.80$, $p=0.005$, $BF=0.677$, $\eta_p^2=0.04$), but no significant interaction ($F(8,323)=0.48$, $p=0.868$, $BF=0.02$, $\eta_p^2=0.01$) supporting the notion that the data across experiments showed the same qualitative pattern.

Supplementary Material

Supplementary Experiment 1

Participants

243 participants (128 female, Age: mean = 30.1, mode = 24, SD = 11.7) were recruited via [prolific.co](#). The hourly compensation was £ 6.3. All participants had normal or corrected-to-normal vision. The sample size was chosen to match that of Experiment 1. The study was approved by the Psychological Research Ethics Board of the Central European University and all participants provided informed consent.

Materials

This experiment used the same materials as Experiment 1. However, the shapes were not grouped into horizontal and vertical pairs but into two orthogonal groups of oblique pairs. For one group of pairs, the second shape was always in the top right grid cell from the first shape, for the other group of pairs the second shape was always in the top-left grid cell. As such pairs allow for fewer unique combinations within a 3×3 grid, the scenes for this Experiment were set in a 5×5 grid. The shapes mainly occupied the central 3×3 sub-grid, with one shape per scene being in the outer cells. All shapes appeared in the outer cells an equal number of times over all scenes. In order to assure that the whole grid is visible on the participants' devices, the size was not set to a fixed pixel value. Instead, the size of one grid cell was set to $1/7$ of the pixel height of the participant's screen. Therefore, the whole grid filled $5/7$ of the screen height.

Procedure

The procedure was identical to Experiment 1.

Results

The exclusion criteria were identical to those used in Experiment 1. This led to 15 exclusions for failed attention checks and 4 exclusions for response bias. This left us with 224 participants after exclusions. Participants were categorized as explicit or implicit in the same way as in Experiment 1. This led to 12 participants being categorized as explicit and 212 as implicit.

Overall, the data shows the same pattern as in Experiment 1 (see [Supplementary Figure 1](#), bottom panel). The results for the implicit participants ($n=212$) closely follow the results of Experiment 1. They perform above chance for pairs of the first training phase ($M=53.9$, $SE=0.8$, $d=0.34$, $t(211)=4.99$ $p<0.001$, $BF=10258$) and for pairs of a novel structure ($M=57.1$, $SE=1.7$, $d=0.28$, $t(211)=4.06$ $p<0.001$, $BF=245$) but not pairs of the same structure ($M=50.1$, $SE=1.8$, $d=0.03$, $t(211)=0.46$ $p=0.999$, $BF=0.12$) in the second training phase. The performance for pairs of the same and novel structure is again significantly different ($d=0.18$, $t(211)=2.67$ $p=0.049$, $BF=3.2$). The results for explicit participants ($n=12$) show the same qualitative pattern as in Experiment 1, however without reaching a significant difference from chance: first training phase: $M=59.4$, $SE=4.8$, $d=0.56$, $t(11)=1.95$ $p=0.387$, $BF=1.2$, novel structure: $M=52.1$, $SE=10.4$, $d=0.06$, $t(11)=0.20$ $p=0.999$, $BF=0.29$, same structure: $M=60.4$, $SE=7.8$, $d=0.39$, $t(11)=1.33$ $p=0.084$, $BF=0.59$). This can be explained by the significantly smaller number of participants acquiring explicitness in Supplementary Experiment 1 as compared to 1 ($X^2=10.47$, $df=1$, $p=0.001$, $BF=37.0$), which results in diminished power for these tests.

Sleep During Consolidation Studies

In order to assure that participants had overnight sleep during Experiments 2 and 4, and that they didn't sleep during the day in Experiment 3, a number of constraints and checks were implemented. First, participants were not taken from the full global prolific pool, but restricted to a number of European countries within the same time zones. Country of residence is one of the attributes of participants that prolific.co verifies. In order to roughly approximate the geographic distribution of participants in Experiment 1 (see [Supplementary Figure 4](#)), we chose countries from two time zones. GMT±00:00 (Countries: UK and Portugal) and GMT+01:00 (Countries: Germany, France, Spain, Czech Republic, Denmark, Hungary, Italy, Netherlands, Poland, Slovenia, Switzerland). Second, as we don't expect Prolific's residence information to be perfectly predictive of where participants are while they conduct the experiment, participants were asked what the current time at their location is. Third, at the start of the second part of the experiment, participants were asked how much they slept between the first and second parts. This was used to exclude participants from Experiment 3 who slept during the day.

Matched Sample Analysis

As reported in the main text, explicit participants show higher average learning in the first learning phase, which could be what enables the generalization of the learned structure. To test this idea, we conducted a matched sample analysis ([39](#)). The general idea of this analysis is to create a sub-sample of the implicit participants that perform like the explicit participants for the pre-training trials.

For all experiments, In a first step, we ran six applicable matching algorithms implemented in the MatchIt R package ([75](#)). The six so-created matched implicit samples were then compared to the original explicit sample according to four metrics: standardized mean difference, variance ratio, mean of the empirical cumulative density function, and maximum of the empirical cumulative density function. All values for all experiments can be seen in Supplementary Tables 1-4. "Unbalanced" denotes the values for the full, non-matched implicit sample. All values for the used matching methods are evaluated as an improvement from those values. Std. Mean Diff. describes how far the mean of the matched sample is from the comparison sample (explicit participants); values closer to zero are better. The variance ratio is the ratio of the variances of the matched and the comparison sample; the best possible value is 1. The eCDF (empirical cumulative density function) contains more information than the mean and variance ratio as they capture the whole distribution of values. Two commonly used simple metrics based on the eCDF are the mean and maximum difference of the eCDFs of the matched and comparison group. Generally, values closer to zero are better. The best-fitting matching algorithm was not exactly aligned for all experiments. For consistency reasons, we chose the overall best-fitting method for all experiments: nearest

neighbour matching with replacement. To provide an detailed overview of how this decision might have influenced our results, we plotted the results for all tested matching algorithms in **Supplementary Figure 5** [↗](#).

A potential shortcoming of the applied matching algorithm is that some participants will reach higher than 50% performance by chance. Therefore, some of the matched implicit participants may not actually have the same level of learning as the explicit participant they are matched to, as they might have achieved their apparent level of performance by chance. However, there are two strong reasons why this is highly unlikely. First, it is obvious from comparing the results from our full sample analysis to the matched sample analysis that the qualitative pattern of results in Phase 2 identical between the two groups, suggesting that Phase 2 performance is virtually independent of Phase 1 performance for implicit learners. This similarity in patterns weakens the assumption that the matched group behaves fundamentally differently from the full implicit group. Second, if participants in the matched implicit group performed better than average only by chance, it is difficult to argue that at the same time their performance across Same and Novel trials had a significantly non-random pattern that was the exact opposite of that shown by the Explicit participants.

Experiment 1 Results

We found that the matched sample showed a similar pattern for learning in the second training phase as the original full sample (see **Figure 3** [↗](#)). This is captured by a 2×2 ANOVA using the novel and same structure pairs for the original explicit and the matched implicit data showing a significant interaction ($F(1,87)=8.53$, $p=0.004$, $BF=10.7$, $\eta_p^2=0.09$) and post-hoc comparisons show a significant difference between novel and same structure trials for the synthetic implicit data ($p=0.012$; $BF=3.6$). This analysis suggests that the difference between the two groups is not merely based on different strengths of learning in the first training phase.

Experiment 2 Results

As in Experiment 1, the matched sample showed the same type of pattern as the full sample (see **Figure 3** [↗](#)). As a critical analysis, we can see that for the matched implicit sample there is a significant difference between learning pairs of the same and of the novel structure ($d=0.98$, $t(20)=4.51$, $p<0.001$, $BF=127$), suggesting generalization of the structure.

Experiment 3 Results

As previously, the matched sample showed a similar type of pattern as the full sample (see **Figure 3** [↗](#)). Critically, we can see that for the matched implicit sample there is no significant difference between learning pairs of the same and of the novel structure ($d=0.29$, $t(19)=-1.29$, $p=0.214$, $BF=0.59$), suggesting no generalization of the structure.

Experiment 4 Results

As in Experiment 1, the matched sample showed the same type of pattern as the full sample descriptively (see **Figure 3** [↗](#)). However, the critical analysis of the difference between learning pairs of the same and of the novel structure for the matched implicit sample failed to reach significance ($M_{diff}=8.66$, $d=0.22$, $t(22)=1.05$, $p=0.304$, $BF=0.46$).

Supplementary Table 1

Experiment 1 - Overview of Balance Metrics for the Used Matching Algorithms

Matching Method	Std. M. Diff.	Var. Ratio	eCDF mean	eCDF max
Unbalanced	0.486	2.92	0.169	0.358
NN with replacement	-0.002	0.95	0.003	0.059
NN without replacement	0.078	1.31	0.026	0.206
Optimal pair matching	0.078	1.31	0.026	0.206
Optimal full matching	-0.018	1.05	0.012	0.059
Coarsened exact matching	-0.021	1.12	0.021	0.118
Subclassification	-0.049	1.19	0.020	0.059

Supplementary Table 2

Experiment 2 - Overview of Balance Metrics for the Used Matching Algorithms

Matching Method	Std. M. Diff.	Var. Ratio	eCDF mean	eCDF max
Unbalanced	1.125	2.909	0.273	0.548
NN with replacement	0.432	2.424	0.105	0.429
NN without replacement	0.103	1.089	0.030	0.238
Optimal pair matching	0.432	2.424	0.105	0.429
Optimal full matching	0.112	1.079	0.033	0.238
Coarsened exact matching	0.142	1.130	0.037	0.294
Subclassification	0.248	1.175	0.060	0.238

Supplementary Table 3

Experiment 3 - Overview of Balance Metrics for the Used Matching Algorithms

Matching Method	Std. M. Diff.	Var. Ratio	eCDF mean	eCDF max
Unbalanced	1.067	3.284	0.290	0.49
NN with replacement	0.293	2.260	0.079	0.30
NN without replacement	0.043	0.950	0.018	0.25
Optimal pair matching	0.293	2.260	0.079	0.30
Optimal full matching	0.068	1.073	0.023	0.25
Coarsened exact matching	0.043	1.161	0.023	0.25
Subclassification	0.089	1.106	0.039	0.25

Supplementary Table 4

Experiment 4 - Overview of Balance Metrics for the Used Matching Algorithms

Matching Method	Std. M. Diff.	Var. Ratio	eCDF mean	eCDF max
Unbalanced	0.819	5.352	0.222	0.494
NN with replacement	0.450	3.006	0.106	0.391
NN without replacement	0.164	1.327	0.043	0.304
Optimal pair matching	0.450	3.006	0.106	0.391
Optimal full matching	0.151	1.479	0.047	0.304
Coarsened exact matching	0.026	1.099	0.016	0.064
Subclassification	0.240	2.404	0.071	0.304

Exit Survey

The test trials of all Experiments were followed by a set of open-ended questions. Two of these questions were jointly used to assess participants' explicitness of knowledge.

Participants were labeled as “explicit” if they gave any indication of being aware of reappearing fixed patterns/pairs in the scenes. Additionally, a small number of participants who clearly reported the underlying structure (horizontality/verticality) were excluded from the analysis.

Questions

1. “Please, explain with your own words what you think the experiment was about.”
2. “In the first part of the experiment, where you only passively watched the screen: Did you notice any regularities in how the shapes were arranged? If yes, please describe them!”

Example Answers used for classification

Reporting Underlying Structure

- “the shapes had consistent pairs in the same row of the grid“
- “There were 5 pairs and all of them were matching vertically.“

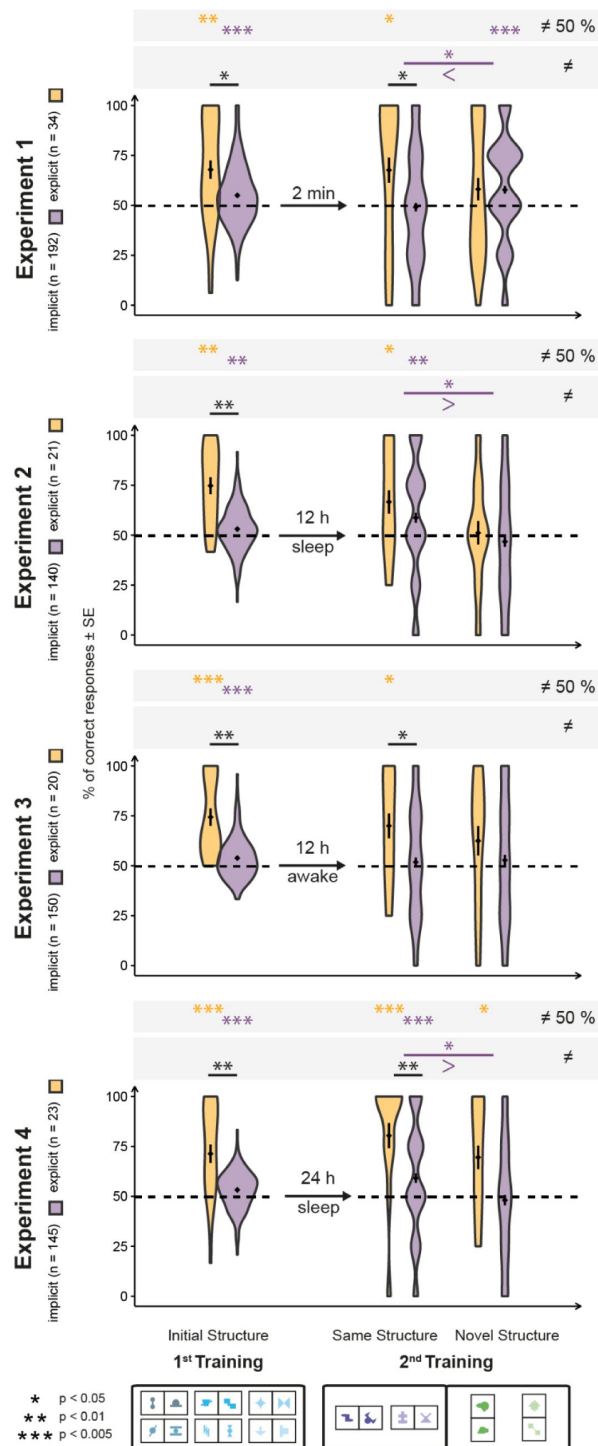
Explicit

- Example of general pair knowledge: “I noticed the same 2 shapes were always together“
- Example of specific pair knowledge: “The ‘flag’ was always paired with the ‘sun’, the ‘square’ with the ‘arrow’, I think.“

Implicit

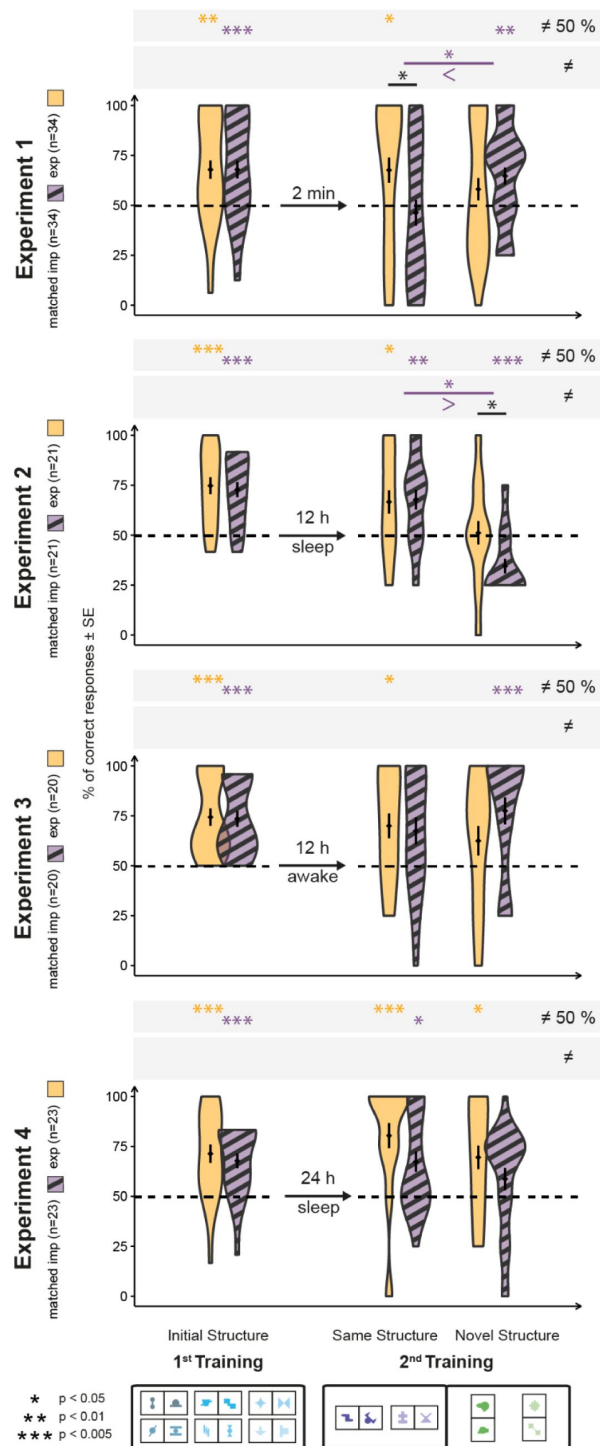
- Example of no reported knowledge: “I tried to find some regularities, but I couldn’t find any. Everything looked random to me“
- Example of no relevant knowledge: “I think that cross symbol was always in the top or bottom row, never in the middle. But I am probably wrong.“

Supplementary Figures



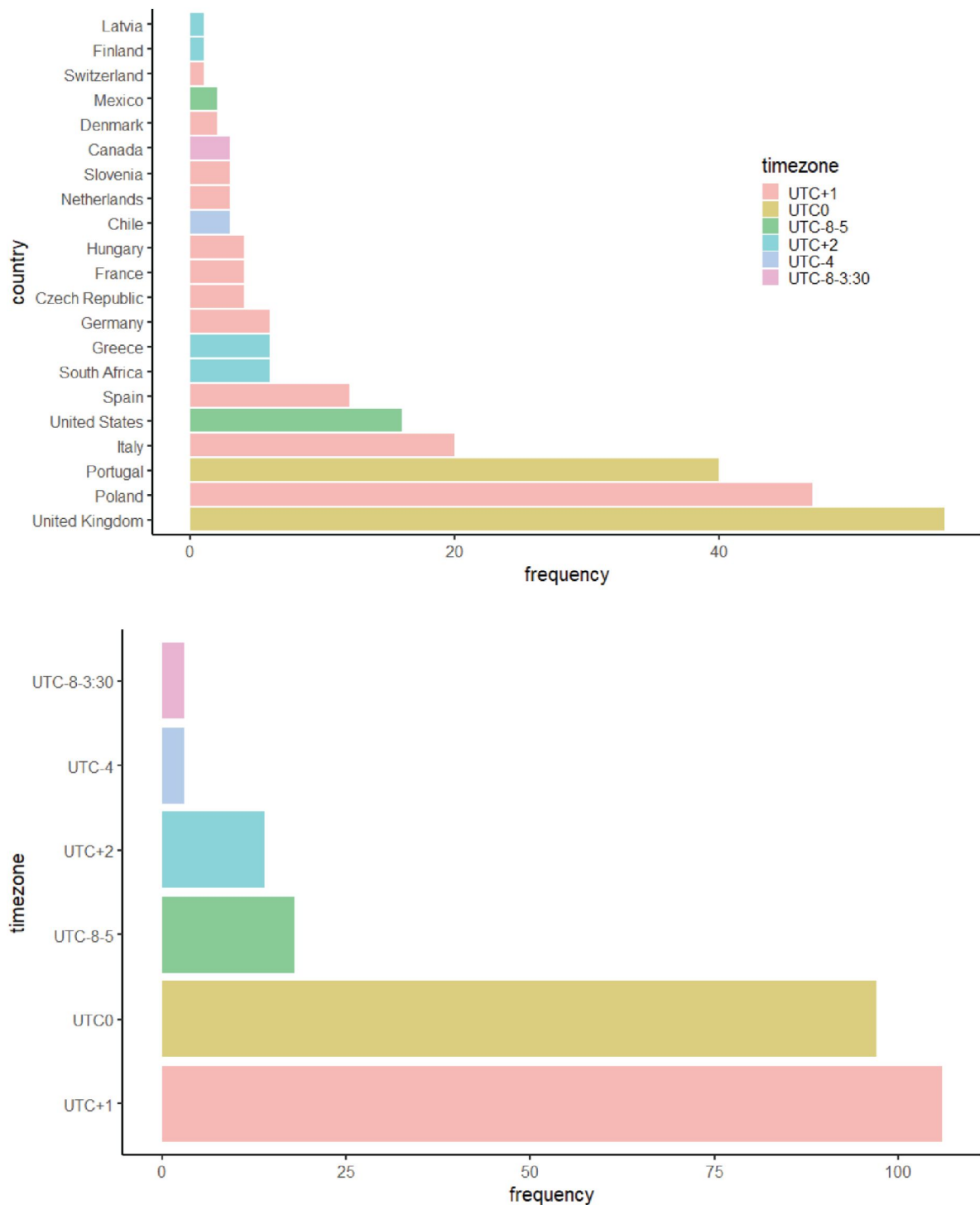
Supplementary Figure 2

Violin plot version of [Figure 2](#), depicting results of familiarity tests in Experiments 1, 2, 3, and 4. The width of the violins corresponds to a smoothed version of the relative frequency of values in the dataset. The markers within the violins denote the mean and standard error. Test results of the 2AFC trials in all four experiments are grouped on the x-axis according to whether the trials used shapes of the 1st or the 2nd training and, within the 2nd training, whether the pair in the trial had the same or different orientation structure as inventory pairs in the 1st training. The y-axis represents the proportion of correct responses in the 2AFC test trials. Arrows and text between the test results related to the two trainings convey the condition and length of the break period. Legend of significance levels is shown in the lower left corner. Signs of inequality below the comparison in the 2nd training indicate the direction of effect.



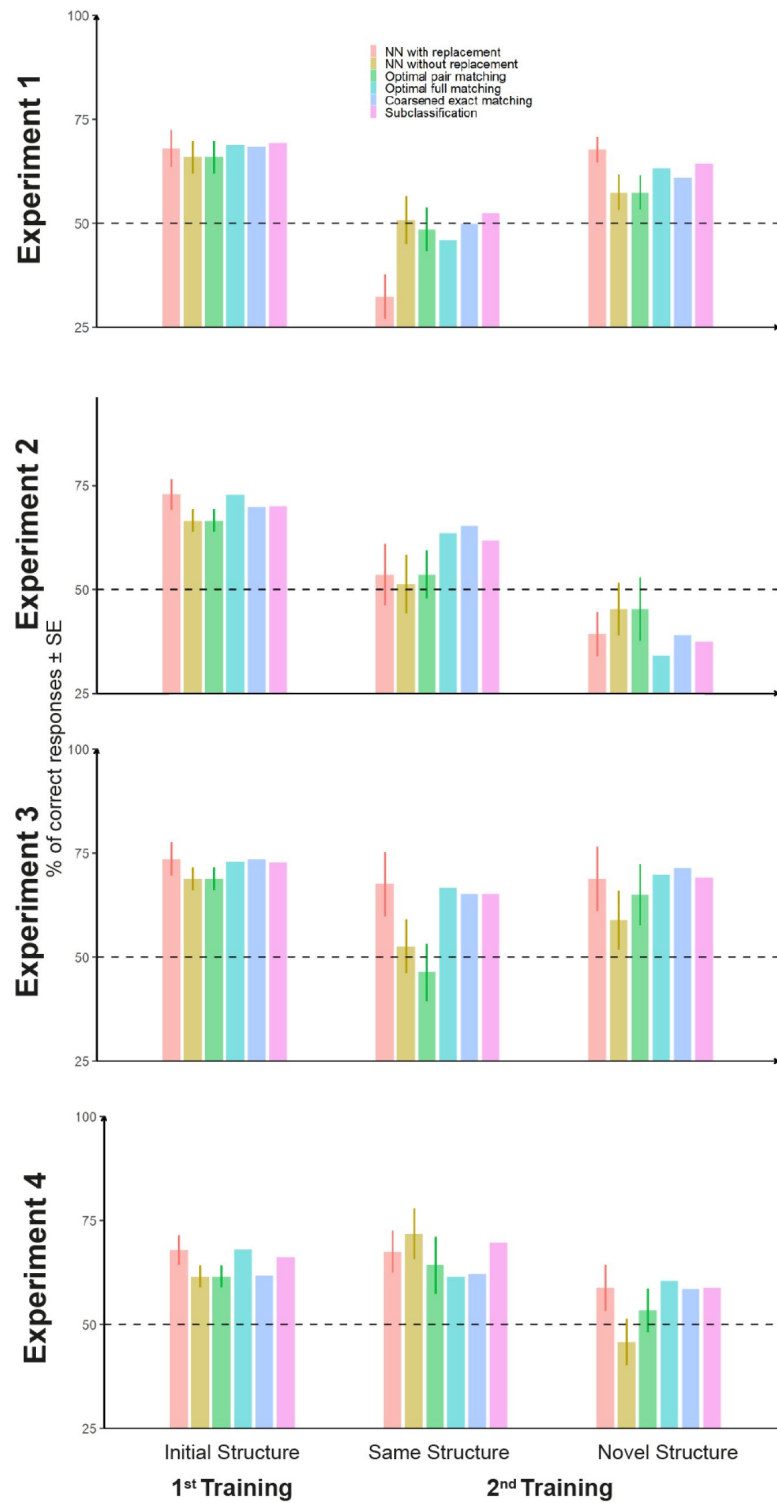
Supplementary Figure 3

Violin plot version of **Figure 3**, depicting results of familiarity tests in the matched data for Experiments 1, 2, 3, and 4. The width of the violins corresponds to a smoothed version of the relative frequency of values in the dataset. The markers within the violins denote the mean and standard error. Test results of the 2AFC trials in all four experiments are grouped on the x-axis according to whether the trials used shapes of the 1st or the 2nd training and, within the 2nd training, whether the pair in the trial had the same or different orientation structure as inventory pairs in the 1st training. The y-axis represents the proportion of correct responses in the 2AFC test trials. Arrows and text between the test results related to the two trainings convey the condition and length of the break period. Legend of significance levels is shown in the lower left corner. Signs of inequality below the comparison in the 2nd training indicate the direction of effect.



Supplementary Figure 4.

Frequency distribution of country of residence and time zone for participants in Experiment 1.



Supplementary Figure 5

Results of familiarity tests in Experiments 1, 2, 3, and 4 for the matched implicit sample plotted for all tested matching algorithms. Test results of the 2AFC trials in both experiments are grouped on the x-axis according to whether the trials used shapes of the 1st or the 2nd training and, within the 2nd training, whether the pair in the trial had the same or different orientation structure as inventory pairs in the 1st training. The y-axis represents the proportion of correct responses in the 2AFC test trials. The horizontal dotted line denotes chance performance. Color coding indicates the used matching algorithm.

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Reviewer #1 (Public review):**Summary:**

This paper investigates the effects of the explicit recognition of statistical structure and sleep consolidation on the transfer of learned structure to novel stimuli. The results show a striking dissociation in transfer ability between explicit and implicit learning of structure, finding that only explicit learners transfer structure immediately. Implicit learners, on the other hand, show an intriguing immediate structural interference effect (better learning of novel structure) followed by successful transfer only after a period of sleep.

Strengths:

This paper is very well written and motivated, and the data are presented clearly with a logical flow. There are several replications and control experiments and analyses that make the pattern of results very compelling. The results are novel and intriguing, providing important constraints on theories of consolidation. The discussion of relevant literature is thorough. In sum, this work makes an exciting and important contribution to the literature.

Weaknesses:

There have been several recent papers which have identified issues with alternative forced choice (AFC) tests as a method of assessing statistical learning (e.g. Isbilen et al. 2020, Cognitive Science). A key argument is that while statistical learning is typically implicit, AFC involves explicit deliberation and therefore does not match the learning process well. The use of AFC in this study thus leaves open the question of whether the AFC measure benefits the explicit learners in particular, given the congruence between knowledge and testing format, and whether, more generally, the results would have been different had the method of assessing generalization been implicit. Prior work has shown that explicit and implicit measures of statistical learning do not always produce the same results (eg. Kiai & Melloni, 2021, bioRxiv; Liu et al. 2023, Cognition).

The authors argued in their response to this point that this issue could have quantitative but not qualitative impacts on the results, but we see no reason that the impact could not be qualitative. In other words, it should be acknowledged that an implicit test could potentially result in the implicit group exhibiting immediate structure transfer.

Given that the explicit/implicit classification was based on an exit survey, it is unclear when participants who are labeled "explicit" gained that explicit knowledge. This might have occurred during or after either of the sessions, which could impact the interpretation of the effects and deserves discussion.

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Reviewer #2 (Public review):**Summary:**

Sleep has not only been shown to support the strengthening of memory traces, but also their transformation. A special form of such transformation is the abstraction of general rules from the presentation of individual exemplars. The current work used large online experiments with hundreds of participants to shed further light on this question. In the training phase participants saw composite items (scenes) that were made up of pairs of spatially coupled (i.e., they were next to each other) abstract shapes. In the initial training, they saw scenes made up of six horizontally structured pairs and in the second training phase, which took place after a retention phase (2 min awake, 12 hour incl. sleep, 12 h only wake, 24 h incl.

sleep), they saw pairs that were horizontally or vertically coupled. After the second training phase, a two-alternatives-forced-choice (2-AFC) paradigm, where participants had to identify true pairs versus randomly assembled foils, was used to measure performance on all pairs. Finally, participants were asked five questions to identify, if they had insight into the pair structure and post-hoc groups were assigned based on this. Mainly the authors find that participants in the 2 minute retention experiment without explicit knowledge of the task structure were at chance level performance for the same structure in the second training phase, but had above chance performance for the vertical structure. The opposite was true for both sleep conditions. In the 12 h wake condition these participants showed no ability to discriminate the pairs from the second training phase at all.

Strengths:

All in all, the study was performed to a high standard and the sample size in the implicit condition was large enough to draw robust conclusions. The authors make several important statistical comparisons and also report an interesting resampling approach. There is also a lot of supplemental data regarding robustness.

Weaknesses:

My main concern regards the small sample size in the explicit group and the lack of experimental control.

<https://doi.org/10.7554/eLife.100785.2.sa2>

Reviewer #3 (Public review):

In this project, Garber and Fiser examined how the structure of incidentally learned regularities influences subsequent learning of regularities, that either have the same structure or a different one. Over a series of six online experiments, it was found that the structure (spatial arrangement) of the first set of regularities affected learning of the second set, indicating that it has indeed been abstracted away from the specific items that have been learned. The effect was found to depend on the explicitness of the original learning: Participants who noticed regularities in the stimuli were better at learning subsequent regularities of the same structure than of a different one. On the other hand, participants whose learning was only implicit had an opposite pattern: they were better in learning regularities of a novel structure than of the same one. However, when an overnight sleep separated the first and second learning phases, this opposite effect was reversed and came to match the pattern of the explicit group, suggesting that the abstraction and transfer in the implicit case were aided by memory consolidation.

In their revision the authors addressed my major comments successfully and I commend them for that.

<https://doi.org/10.7554/eLife.100785.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

This paper investigates the effects of the explicit recognition of statistical structure and sleep consolidation on the transfer of learned structure to novel stimuli. The results show a striking dissociation in transfer ability between explicit and implicit learning of structure, finding that only explicit learners transfer structure immediately. Implicit learners, on the other hand, show an intriguing immediate structural interference effect (better learning of novel structure) followed by successful transfer only after a period of sleep.

Strengths:

This paper is very well written and motivated, and the data are presented clearly with a logical flow. There are several replications and control experiments and analyses that make the pattern of results very compelling. The results are novel and intriguing, providing important constraints on theories of consolidation. The discussion of relevant literature is thorough. In summary, this work makes an exciting and important contribution to the literature.

Weaknesses:

There have been several recent papers that have identified issues with alternative forced choice (AFC) tests as a method of assessing statistical learning (e.g. Isbilen et al. 2020, Cognitive Science). A key argument is that while statistical learning is typically implicit, AFC involves explicit deliberation and therefore does not match the learning process well. The use of AFC in this study thus leaves open the question of whether the AFC measure benefits the explicit learners in particular, given the congruence between knowledge and testing format, and whether, more generally, the results would have been different had the method of assessing generalization been implicit. Prior work has shown that explicit and implicit measures of statistical learning do not always produce the same results (eg. Kiai & Melloni, 2021, bioRxiv; Liu et al. 2023, Cognition).

We agree that numerous papers in the Statistical Learning literature discuss how different test measures can lead to different results and, in principle, using a different measure could have led to varying results in our study. In addition, we believe there are numerous additional factors relevant to this issue including the dichotomous vs. continuous nature of implicit vs. explicit learning and the complexity of the interactions between the (degree of) explicitness of the participants' knowledge and the applied test method that transcend a simple labeling of tests as implicit or explicit and that strongly constrains the type of variations the results of different test would produce. Therefore, running the same experiments with different learning measures in future studies could provide additional interesting data with potentially different results.

However, the most important aspect of our reply concerning the reviewer's comment is that although quantitative differences between the learning rate of explicit and implicit learners are reported in our study, they are not of central importance to our interpretations. What is central are the different qualitative patterns of performance shown by the explicit and the implicit learners, i.e., the opposite directions of learning differences for “novel” and “same” structure pairs, which are seen in comparisons within the explicit group vs. within the implicit group and in the reported interaction. Following the reviewer's concern, any advantage an explicit participant might have in responding to 2AFC trials using “novel” structure pairs should also be present in the replies of 2AFC trials using the “same” structure pairs and this effect, at best, could modulate the overall magnitude of the across groups (Expl/Impl.) effect but not the relative magnitudes within one group. Therefore, we see no parsimonious reason to believe that any additional interaction between the explicitness level of participants and the chosen test type would impede our results and their interpretation.

Given that the explicit/implicit classification was based on an exit survey, it is unclear when participants who are labeled "explicit" gained that explicit knowledge. This might have occurred during or after either of the sessions, which could impact the interpretation of the effects.

We agree that this is a shortcoming of the current design, and obtaining the information about participants' learning immediately after Phase 1 would have been preferred. However, we made this choice deliberately as the disadvantage of assessing the level of learning at the end of the experiment is far less damaging than the alternative of exposing the participants to the exit survey question earlier and thereby letting them achieve explicitness or influence their mindset otherwise through contemplating the survey questions before Phase 2. Our Experiment 5 shows how realistic this danger of unwanted influence is: with a single sentence alluding to pairs in the instructions of Exp 5, we could completely change participants' quantitative performance and qualitative response pattern. Unfortunately, there is no implicit assessment of explicitness we could use in our experimental setup. We also note that given the cumulative nature of statistical learning, we expect that the effect of using an exit survey for this assessment only shifts absolute magnitudes (i.e. the fraction of people who would fall into the explicit vs. implicit groups) but not aspects of the results that would influence our conclusions.

Reviewer #2 (Public Review):

Summary:

Sleep has not only been shown to support the strengthening of memory traces but also their transformation. A special form of such transformation is the abstraction of general rules from the presentation of individual exemplars. The current work used large online experiments with hundreds of participants to shed further light on this question. In the training phase, participants saw composite items (scenes) that were made up of pairs of spatially coupled (i.e., they were next to each other) abstract shapes. In the initial training, they saw scenes made up of six horizontally structured pairs, and in the second training phase, which took place after a retention phase (2 min awake, 12 h incl. sleep, 12 h only wake, 24 h incl. sleep), they saw pairs that were horizontally or vertically coupled. After the second training phase, a two-alternatives-forced-choice (2-AFC) paradigm, where participants had to identify true pairs versus randomly assembled foils, was used to measure the performance of all pairs. Finally, participants were asked five questions to identify, if they had insight into the pair structure, and post-hoc groups were assigned based on this. Mainly the authors find that participants in the 2-minute retention experiment without explicit knowledge of the task structure were at chance level performance for the same structure in the second training phase, but had above chance performance for the vertical structure. The opposite was true for both sleep conditions. In the 12 h wake condition these participants showed no ability to discriminate the pairs from the second training phase at all.

Strengths:

All in all, the study was performed to a high standard and the sample size in the implicit condition was large enough to draw robust conclusions. The authors make several important statistical comparisons and also report an interesting resampling approach. There is also a lot of supplemental data regarding robustness.

Weaknesses:

My main concern regards the small sample size in the explicit group and the lack of experimental control.

The sample sizes of the explicit participants in our experiments are, indeed, much smaller than those of the implicit participants due to the process of how we obtain the members of the two groups. However, these sample sizes of the explicit groups are not small at all compared to typical experiments reported in Visual Statistical Learning studies, rather they tend to be average to large sizes. It is the sizes of the implicit subgroups that are unusually high due to the aforementioned data collecting process. Moreover, the explicit subgroups have significantly larger effect sizes than the implicit subgroup, bolstering the achieved power that is also confirmed by the reported Bayes Factors that support the “effect” or the “no effect” conclusions in the various tests ranging in value from substantial to very strong. Based on these statistical measures, we think the sample sizes of the explicit participants in our studies are adequate.

As for the lack of experimental control, indeed, we could not fully randomize consolidation condition assignment. Instead, the assignment was a product of when the study was made available on the online platform Prolific. This method could, in theory, lead to an unobserved covariate, such as morningness, being unbalanced between conditions. We do not have any reasons to believe that such a condition would critically alter the effects reported in our study, but as it follows from the nature of unobserved variables, we obviously cannot state this with certainty. Therefore, we added an explicit discussion of these potential pitfalls in the revised version of the manuscript.

Reviewer #3 (Public Review):

In this project, Garber and Fiser examined how the structure of incidentally learned regularities influences subsequent learning of regularities, that either have the same structure or a different one. Over a series of six online experiments, it was found that the structure (spatial arrangement) of the first set of regularities affected the learning of the second set, indicating that it has indeed been abstracted away from the specific items that have been learned. The effect was found to depend on the explicitness of the original learning: Participants who noticed regularities in the stimuli were better at learning subsequent regularities of the same structure than of a different one. On the other hand, participants whose learning was only implicit had an opposite pattern: they were better in learning regularities of a novel structure than of the same one. This opposite effect was reversed and came to match the pattern of the explicit group when an overnight sleep separated the first and second learning phases, suggesting that the abstraction and transfer in the implicit case were aided by memory consolidation.

These results are interesting and can bridge several open gaps between different areas of study in learning and memory. However, I feel that a few issues in the manuscript need addressing for the results to be completely convincing:

(1) The reported studies have a wonderful and complex design. The complexity is warranted, as it aims to address several questions at once, and the data is robust enough to support such an endeavor. However, this work would benefit from more statistical rigor. First, the authors base their results on multiple t-tests conducted on different variables in the data. Analysis of a complex design should begin with a large model incorporating all variables of interest. Only then, significant findings would warrant further follow-up investigation into simple effects (e.g., first find an interaction effect between group and novelty, and only then dive into what drives that interaction). Furthermore, regardless of the statistical strategy used, a correction for multiple comparisons is needed here. Otherwise, it is hard to be convinced that none of these effects are spurious. Last, there is considerable variation in sample size between experiments. As the authors have conducted a power analysis, it would be good to report that information per each experiment, so readers know what power to expect in each.

Answering the questions we were interested in required us to investigate two related but separate types of effects within our data: general above-chance performance in learning, and within- and across-group differences.

Above-chance performance: As typical in SL studies, we needed to assess whether learning happened at all and which types of items were learned. For this, a comparison to the chance level is crucial and, therefore, one-sample t-test is the statistical test of choice. Note that all our t-tests were subject to experiment-wise correction for multiple comparisons using the Holm-Bonferroni procedure, as reported in the Supplementary Materials.

Within- and across-group differences: To obtain our results regarding group and par-type differences and their interactions, we used mixed ANOVAs and appropriate post-hoc tests as the reviewer suggested. These results are reported in the method section.

Concerning power analysis, in the revised version of the manuscript we added analysis of achieved power for the statistical tests most critical to our arguments.

(2) Some methodological details in this manuscript I found murky, which makes it hard to interpret results. For example, the secondary results section of Exp1 (under Methods) states that phase 2 foils for one structure were made of items of the other structure. This is an important detail, as it may make testing in phase 2 easier, and tie learning of one structure to the other. As a result, the authors infer a "consistency effect", and only 8 test trials are said to be used in all subsequent analyses of all experiments. I found the details, interpretation, and decision in this paragraph to lack sufficient detail, justification, and visibility. I could not find either of these important design and analysis decisions reflected in the main text of the manuscript or in the design figure. I would also expect to see a report of results when using all the data as originally planned.

We thank the reviewer for pointing out these critical open questions our manuscript that need further clarification. The inferred "consistency effect" is based on patterns found in the data, which show an increase in negative correlation between test types during the test phase. As this is apparently an effect of the design of the test phase and not an effect of the training phase, which we were interested in, we decided to minimize this effect as far as possible by focusing on the early test trials. For the revised version of the manuscript, we revamped and expanded the discussion of how this issue was handled and also add a short comment in the main text, mentioning the use of only a subset of test trials and pointing the interested reader to the details.

Similarly, the matched sample analysis is a great addition, but details are missing. Most importantly, it was not clear to me why the same matching method should be used for all experiments instead of choosing the best matching subgroup (regardless of how it was arrived at), and why the nearest-neighbor method with replacement was chosen, as it is not evident from the numbers in Supplementary Table 1 that it was indeed the best-performing method overall. Such omissions hinder interpreting the work.

Since our approach provided four different balanced metrics (see Supp. Tables 1-4) for each matching method, it is not completely straightforward to make a principled decision across the methods. In addition, selecting the best method for each experiment separately carries the suspicion of cherry-picking the most suitable results for our purposes. For the revised version, we expanded on our description of the matching and decision process and added supplementary descriptive plots showing what our data looks like under each matching method for each experiment. These plots highlight that the matching techniques produce qualitatively roughly identical results and picking one of them over the other does not alter

the conclusions of the test. The plots give the interested reader all the necessary information to assess the extent our design decisions influence our results.

(3) To me, the most surprising result in this work relates to the performance of implicit participants when phase 2 followed phase 1 almost immediately (Experiment 1 and Supplementary Experiment 1). These participants had a deficit in learning the same structure but a benefit in learning the novel one. The first part is easier to reconcile, as primacy effects have been reported in statistical learning literature, and so new learning in this second phase could be expected to be worse. However, a simultaneous benefit in learning pairs of a new structure ("structural novelty effect") is harder to explain, and I could not find a satisfactory explanation in the manuscript.

Although we might not have worded it clearly, we do not claim that our "structural novelty effect" comes from a "benefit" in learning pairs of the novel structure. Rather, we used the term "interference" and lack of this interference. In other words, we believe that one possible explanation is that there is no actual benefit for learning pairs of the novel structure but simply unhindered learning for pairs of the novel structure and simultaneous inference for learning pairs of the same structure. Stronger interference for the same compared to the novel structure items seems as a reasonable interpretation as similarity-based interference is well established in the general (not SL-specific) literature under the label of proactive interference.

After possible design and statistical confounds (my previous comments) are ruled out, a deeper treatment of this finding would be warranted, both empirically (e.g., do explicit participants collapse across Experiments 1 and Supplementary Experiment 1 show the same effect?) and theoretically (e.g., why would this phenomenon be unique only to implicit learning, and why would it dissipate after a long awake break?).

Across all experiments, the explicit participants showed the same pattern of results but no significant difference between pair types, probably due to insufficiency of the available sample sizes. We already included in the main text the collapsed explicit results across Experiments 1-4 and Supplementary Experiment 1 (p. 16). This analysis confirmed that, indeed, there was a significant generalization for explicit participants across the two learning phases. We could re-run the same analysis for only Experiment 1 and Supplementary Experiment 1, but due to the small sample of $N=12$ in Suppl. Exp. 1, this test will be likely completely underpowered. Obtaining the sufficient sample size for this one test would require an excessive number (several hundreds) of new participants.

In terms of theoretical treatment, we already presented our interpretation of our results in the discussion section, which we expanded on in the revised manuscript.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) It would be very useful to add individual data points (and/or another depiction of the distribution) to the bar plots. If not in the main figures, as added figures in the supplement.

We added violin plots for all results in the Supplementary.

(2) It would be helpful to include in the supplement some examples of responses that led to the 'explicit' or 'implicit' classification. Specifically, what kind of response was considered to contain a partial recognition of the underlying structure vs. no recognition?

We added example responses used for classification in the Supplementary.

(3) It would be useful to show the results of Experiment 5 as well as the diagonal version as supplemental figures.

We added the requested figures in the Supplementary.

Typos: page 10: "in in the tests", page 15: "rerun"

Fixed.

Reviewer #2 (Recommendations For The Authors):

(1) My strongest reservation relates to the small sample size in the explicit group. The authors do report stats for all experiments together in one analysis and I think this is the only robust finding for this group. I would suggest removing any comparisons between this smaller group and the larger implicit group since they do not make a lot of sense due to the imbalance in sample size in my opinion. If they do want to report the explicit group individually for each experiment, they should at least test for differences between the experiments also for this group using ANOVA.

We do agree that the unbalanced nature of the sample sizes can be problematic for the between-group comparisons. The t-tests reported for between-group comparisons are in fact Welch's t-test better suited for unequal sample sizes and variances. Previously, we failed to report that these t-tests were Welch's t-test, which we fixed in the revised version.

In the Supplementary, we previously reported an ANOVA including all explicit participants from all experiments. This showed a significant main effect of Experiment and test type, but no significant interaction. We take this as evidence that although specific levels of learning vary by experimental condition, the overall pattern of learning (i.e. which pairs are learned better) are the same across all experiments.

(2) Moreover, the explicit group does not only differ in the explicitness of their memory but also regarding learning performance per se (as evidenced by performance differences for the first training). This important confound needs to be acknowledged and discussed more thoroughly!

We agree that this topic is important, this is why the subsection "The Type of Transfer Depends on Quality of Knowledge, Not Quantity of Knowledge" deals exclusively with this issue. See our reply to the next point.

(3) The resampling approach is somewhat interesting to solve the issue raised in 2. However, I doubt that the authors actually achieve what they are claiming. Since we have a 2-AFC task the possibility must be considered that participants who chose correctly in the implicit group did so by chance. This means that the assumption that the matched pairs actually have the same amount of memory for the first training period as the explicit group is likely false. Therefore, this analysis is still comparing apples and oranges.

We address this idea in detail in the supplementary materials pointing out first that the matched results showed the same pattern as the full results suggesting that Phase 1 and Phase 2 results are independent for this group, and by arguing that randomly selected subset of participants should not show a significant deviation from null performance in the Same vs. Novel performance in Phase 2.

(4) One important issue, when conducting online experiments is assuring random allocation of participants. How did the authors recruit participants to ensure they did not select participants for the different experiments that differed regarding their preference for wake vs. sleep retention intervals? If no care was taken in this regard, I would suggest reporting this and maybe briefly discussing it.

This shortcoming was now reported and addressed in the discussion section of the revised manuscript.

(5) I could not find any information about the exact questions that were asked about the task rules. Also, there was no information on how the answers were used to assign groups. Both should be added.

The exact questions were added to the revised Supplementary.

(6) I think that the literature on sleep and rule extraction is well-represented in the manuscript. However, I think also referring more thoroughly to the literature on how sleep leads to gist extraction, schemas, and insight would help understand the relevance of the present research.

We subsumed references to the mentioned areas of research under the labels of abstraction and generalization. In the revised section, we listed the appropriate labels along with the already used references to make the connection to a vast literature treating generalization in related but distinct ways more explicit.

(7) It is unclear to me why the items learned in the first learning phase interfere with those learned in the second learning phase (without sleep) and not vice versa. What is the author's explanation for this?

We added a paragraph on this to our revised discussion section. In short, there may also be retroactive interference. However, we would need yet another variation of the paradigm to properly measure it, and this was outside the scope of the current work.

(8) As far as I can tell the study lacks all of the usual control tasks that are used in the field of sleep and memory (especially subjective sleepiness and objective vigilance). In addition, this research has the circadian confound, and therefore additional controls would have been warranted, e.g., morningness-eveningness, retrieval capabilities. Also, performance immediately after training phase 1 was not tested, which would serve as an important control for circadian differences in initial learning of the rule.

The study uses a number of the control measures established in the sleep and memory literature, such as habitual sleep quality and sleep quality during the night of and the night before the experiment. However, there are, of course, more potentially interesting measures, such as the ones named by the reviewer.

Testing performance right after training phase 1 would have been very interesting indeed. However, due to the nature of statistical learning tasks, this would have completely confounded the implicitness of learning by presenting participants with segmented input; i.e. isolated pairs. Therefore, we opted for the lesser of two evils in our design decision.

(9) As far as I can tell, there is no effect of sleep on correctly identifying pairs from training phase 1. This would be expected and thus should be discussed.

As noted and referenced in the discussion section, the effect of sleep on statistical learning per se is a subject of controversy in the literature, where some studies apparently find effects, while others find no effect on statistical learning whatsoever.

(10) *The manuscript should explicitly mention if the study was preregistered.*

It was not.

Reviewer #3 (Recommendations For The Authors):

The topic of this project is close to my heart, and I commend the authors for conducting numerous variations of the experiment with large sample sizes. I have some suggestions I feel will make the paper stronger, and a few minor comments that caught my eye during reading:

(1) First and foremost, I found the paper's structure cumbersome. For instance, different aspects of Experiment 1 results are reported in (1) the main text, (2) under methods, and (3) in Supplementary. This makes reading unnecessarily difficult. This relates not only to the analysis results - the sample size is reported as 226 in the main text, 226+3 in Methods, and 226+3+19 in Supplementary. I strongly suggest removing all results from the Methods section and merging the supplementary results with the main results.

We overhauled the structure of the paper, moving much more information into the proper method section and out of the Supplementary.

(2) "Attention checks" and "response bias" appear first in Supplementary Experiment 1 but are explained only later under Experiment 1. The same thing for the experimental procedure. I therefore suggest placing Experiment 1 before Supplementary Experiment 1, but related to my previous comment - have one paragraph dedicated to Subject Exclusion of all experiments.

The new structure of the Method sections solves this.

(3) Figure 4 is mentioned but does not appear in the manuscript.

This has been fixed. The paragraph in question now references the correct supplementary figure.

(4) OSF project includes only data with no README file on how to understand the data. The work would also benefit from sharing the experimental and analysis codes.

A README file was added.

(5) This sentence is repeated in relation to four experiments: "Bayes Factors from Bayesian t-tests for implicit participants reported for experiments 1, 2, and 3 used an r-scale parameter of 0.5 instead of the default $\sqrt{2}/2$, reflecting that Experiment 1 found small effect sizes for this group". First, it is missing an explanation of what the r-scale means. Second, it sounds as if this was a product of the procedure, but in fact it was a decision by the researcher if I am correct. If so, it is missing a description of how and why this choice was made.

This was indeed a decision by the researchers, in line with a Bayesian logic of evidence accumulation. We made the explanation in the paper clearer.

(6) Did I understand correctly that each pair was tested 4 times? Was it against the same foil? Did you make sure not to repeat the same pair in back-to-back trials? These details, in addition to what I noted in the public review, are needed.

Each pair was tested 4 times. Each time against a different foil pair. Details have been added to the Method section.

(7) Also in relation to my public review, I could not understand why the sample size was overshoot by so much in Experiment 1 (229 instead of 198.15)?

The calculated sample size of 198.15 was for the implicit subgroup alone, while 229 included explicit and implicit participants.

(8) The correlation between phase 1 and phase 2 is only tested in explicit participants. Why is that? A test in implicit participants is needed for completeness.

Correlations for implicit participants have been added.

(9) There is known asymmetry between the horizontal and vertical plains in our visual system (with preference for horizontal stimuli). I was missing a comparison between learning in the two structures, and a report of how many participants received either in Phase 1.

The allocation of participants to horizontal and vertical conditions was balanced. In the Method section we already report an ANOVA testing for a potential effect of orientation condition, which was not significant.

Minor/aesthetic comments:

(1) "In Phase 2, explicit participants performed above chance for learning pairs that shared their higher level orientation structure with that of pairs in Phase 1". This sounds as if there was a separate test following the two learning phases. Perhaps reword to "for phase 2 pairs".

Fixed

(2) "the two asleep-consolidation groups (Exp. 3 and 4)" - I think you mean Exp. 2 and 4.

Fixed.

(3) "acquiring explicitness in Experiment 5 as compared to 1" I think you mean Supplementary Experiment 1 as compared to 1.

Fixed

(4) "without such a redescription, the previously learned patterns in Phase 1 interfere with new ones in Phase 2, when redescription occurs..." The comma should be a dot.

Fixed

(5) In Experiment 4, did 168 or 169 participants survive exclusion? Both accounts exist, and so do reports of degrees of freedom that allow both 23 and 24 explicit participants.

Fixed.

| (6) *"Implicit learners also performed above chance.." in Experiment 2 is missing (n=XX).*

Fixed.

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